

The Effects of Four-Day School Weeks in Texas

Gaby Lohner* and Justine Weng[†]

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Abstract

Four-day school weeks (FDSWs) have expanded rapidly in Texas in recent years. Districts cite teacher retention and recruitment as the primary motivation, though little is known about the impacts of FDSWs in this context. We study the effects of FDSW adoption on students and teachers in Texas using a staggered differences-in-differences design with a propensity score matched control group. We find that adopting a FDSW leads to positive effects on teacher retention. Teachers in districts that adopt FDSWs are 2.1 percentage points more likely to remain in that district the following year, driven by a decrease in moving to other districts rather than leaving teaching all together. Additionally, adopting a FDSW leads to recruiting higher quality teachers, as measured by having higher value-added scores, more years of experience, and being more likely to hold a Master's degree. In terms of students, we do not find evidence of negative effects on standardized test scores in either math or reading. However, we find heterogeneity in these effects both by policy design and district quality. We find negative effects on student achievement for districts that had the fewest instructional minutes after adoption and for those that had low accountability scores prior to adoption, which we use to proxy for district quality. These results indicate that while the adoption of a FDSW can lead to modest gains in teacher retention, the context in which the policy is adopted and how districts choose to structure the four-day week are important for effects on student achievement.

*Goldman School of Public Policy, UC Berkeley (gaby.lohner@berkeley.edu)

[†]Department of Economics, UC Berkeley (justineweng@berkeley.edu).

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1 Introduction

While much recent attention has been given to the idea of a four-day work week, there has been a quieter, gradual move to four-day school weeks across the country. The concept of a four-day school week (henceforth, FDSW) is straightforward: reducing the school week from the usual five days to four. There has been an increase in the number of school districts across the United States that have switched to this model; as of 2022, over 900 districts nationally were operating on a four-day calendar. While this is less than 10% of all districts, there has been rapid growth in FDSW adoption; from 2019 to 2022, FDSW adoption increased by over 30% (Hollingsworth, 2023). This is especially true in the state of Texas, where there has been a recent wave of FDSW adoption.

In 2015, Texas passed legislation that changed the requirement for school calendars to meet a minimum number of instructional days to instructional minutes. This change allowed districts to concentrate the required instructional minutes in four days instead of five, making a four-day school week possible. The first district in Texas to fully implement a FDSW did so in the 2019-20 academic year (AY2020) and nearly 180 districts in Texas have implemented some version of a FDSW through 2025.¹ Many of these districts cite teacher retention and recruitment as motivating factors for implementing this policy.

This paper examines the effects of FDSW adoption in Texas. We employ a staggered differences-in-differences design to measure the causal impact of FDSWs on both student and teacher outcomes. For teachers, our results show that FDSW adoption leads to an increase in teacher retention, with the share of teachers who stay in their district rising by 2.1 percentage points, driven primarily by a decrease in the share of teachers who move to another district. We also find that new hires following FDSW adoption tend to have more years of experience, are more likely to hold a Master’s degree, and have higher pre-policy value-added scores, suggesting that FDSW adoption allows districts to hire higher quality candidates.

On average, we find that FDSW adoption has limited effects on student achievement, with no changes in math or reading standardized test scores. In addition to standardized tests, we investigate impacts on student behavioral outcomes such as attendance, discipline, and work behavior, as well as medium-run outcomes such as graduation and college enrollment. On average, we find no impacts of the FDSW on these dimensions, though our event study estimates on work behavior and college-going and show evidence of growing effects in the years following adoption.

Though we find no impacts on standardized test scores overall, these results mask substantial heterogeneity depending on the conditions under which a district adopts a FDSW. First, we find that effects differ sharply by how much instructional time districts preserve under the shortened week. Districts that cut the most instructional minutes see declines in both math and reading achievement, while districts that extend the remaining four days to preserve instructional time see minimal reductions in student performance and, in some cases, modest gains. Second, we find that a district’s baseline conditions prior to adoption

¹Prior to 2019-20, two districts had adopted a hybrid four-day school week, having some combination of full five-day weeks and four-day weeks.

matter: districts with low accountability ratings prior to FDSW adoption, which we use to proxy for district quality, see decreases in achievement, while districts with stronger accountability ratings see no negative effects.

Our results contribute to the growing body of empirical literature investigating the impact of four-day school weeks. We are among the first to extend this literature to the context of Texas, where the newest wave of FDSW adoption is occurring, and the first to study the impact on students in Texas. There is mixed evidence on the impact of FDSWs on teacher retention, an important rationale for this policy in Texas. Recent work in Arkansas (Camp, 2024) and Texas (Khalid, 2025; Lawson, 2025) finds evidence of increased teacher retention at schools that adopted a FDSW. On the other hand, Thompson (2025) examines the impact of FDSWs in Missouri and Oklahoma and finds that FDSW adoption has no overall effect on teacher retention and small positive effects among teachers who would have otherwise moved to a FDSW district. Studies on the impact of FDSWs on students in other states have found that the policy led to decreased math and reading achievement for students (Thompson, 2021b; Thompson et al., 2022), though estimates are often imprecise. Our paper contributes to this literature by showing that, on average, FDSWs have a precise null effect on student achievement in Texas. Additionally, our heterogeneity analysis provides evidence on how the effects of FDSWs may depend on the context in which the policy is adopted.

The paper proceeds as follows. Section 2 provides additional background on four-day school weeks and reviews the literature on their impacts. In Section 3, we describe the data used in this study and define our main outcomes. Section 4 details our empirical strategy. In Section 5, we present our main findings on student and teacher outcomes and investigate various factors driving our results. Section 6 concludes.

2 Background

The idea of a four-day school week has been around for nearly a century, potentially as early as 1931 in South Dakota and the 1970s in New Mexico. However, it wasn't until the mid-2000s that the idea gained popularity and we began to see wider adoption in Colorado and Oregon (Hewitt and Denny, 2011). Since then, their prevalence has spread across the country, especially in Western states. Despite potential concerns around how FDSWs may impact student achievement or childcare needs, surveys of school stakeholders, including parents, show positive sentiment toward FDSWs. Over 75% of parents and 85% of students report high satisfaction with their current FDSW in a recent RAND survey (Kilburn et al., 2021).

FDSWs are feasible in states that have a minimum instructional minutes requirement as opposed to instructional days, since schools can compress the necessary minutes into a four-day week schedule. Thompson et al. (2021) compile a comprehensive dataset of FDSWs and analyze implementation across these districts. They find that, as expected, FDSWs have one less instructional day per week, often a Monday or Friday, but the remaining days are, on average, one hour longer than in schools with five-day weeks. Even with the longer days, FDSW districts have fewer total instructional minutes in the academic year. FDSW dis-

districts tend to be more rural, smaller, and have a higher share of White students. Initially, districts' rationale for adopting a FDSW was mainly financial as FDSWs were identified as a potential cost saving mechanism after the Great Recession². Other reasons given for FDSW adoption include attendance issues, teacher shortages, long commutes, and farming commitments, as these are primarily rural areas. Below, we describe the adoption of FDSWs in Texas.

2.1 FDSWs in Texas

In 2015, Texas passed House Bill 2610, which replaced the minimum instructional days requirement with a minimum instructional *minutes* requirement. The cited reason for this change was district flexibility in cases of closures such as inclement weather.³ This bill made four-day school weeks possible in Texas by allowing districts to concentrate instructional minutes in fewer instructional days. District officials in Texas cite the need to retain and recruit teachers as their primary justification for adopting FDSWs.⁴ Additional justifications include student attendance, cost saving, and student mental health.

Priyanti et al. (2024) provides an overview and thematic analysis of the adoption of FDSWs in Texas. The authors find that the majority of FDSW districts in Texas use Friday as their day off, so schools run Monday through Thursday. School days in FDSW districts are 50 to 110 minutes longer than in traditional five-day districts. The authors also find that the vast majority of FDSWs in Texas use the fifth day as a teacher workday or professional development day. This means teachers are not necessarily working on a four-day work week but may have flexible time on the fifth day for activities like lesson planning or grading as there is no student instruction. Finally, the authors note that over 80% of FDSW districts offer no additional academic support for students, such as tutoring or remediation for struggling students. These descriptive patterns highlight that the implementation of FDSWs in Texas largely aligns with what has been done in other states, as discussed in Thompson et al. (2021), with the main difference being that FDSW districts in Texas have slightly longer school days.

One important consequence of adopting a FDSW is that students are home for an extra day during the week, potentially needing additional childcare. To better understand the environment of the districts that adopt FDSWs, we use the American Community Survey - Education Tabulation (ACS-ED) data to present economic characteristics such as workforce participation and industry of employment among parents in FDSW versus traditional districts. The ACS-ED defines "parents of relevant children" as parents of children enrolled in public school; we take these data to provide a sense of the work behavior of students' parents in our sample. We use the 2015-2019 vintage of this data to describe these areas prior to the adoption of FDSWs. As can be seen in the first panel of Table 2, in general, labor force participation rates are roughly similar between parents at FDSW and traditional districts (80.5% vs 81%), as is female labor force participation (70.2% vs 70.9%). The unemployment rate among parents is around 3.5% for both types of dis-

²As discussed below, there is not empirical evidence to support this

³See [this](#) committee report.

⁴See [this](#) news article.

districts. FDSW districts have significantly lower median earnings, \$37,567 compared to \$41,761 at traditional districts. Finally, the industry mix of employment is similar for parents in FDSW districts and traditional districts, though parents in FDSW districts are around 1.6 percentage points more likely to work in agriculture. Overall, this comparison highlights that while there are some differences in the economic context in which a FDSW is adopted, these differences are small. It does not appear, for example, that these areas have lower labor force participation among parents, which might have eased the childcare burden associated with a four-day schedule, or that parents in FDSW districts seem to be working in very different industries than those in districts that decide not to adopt a FDSW.

In a report from the Texas Education Agency (TEA), FDSW districts are compared to a propensity-score matched sample of control districts. The authors find that students in FDSW districts performed worse on standardized math and reading tests (Texas Education Agency, 2025). This work, however, is descriptive and does not account for pre-trends in the outcomes of interest across these two types of districts. To our knowledge, there are two working papers that investigate the causal effect of four-day school weeks in Texas, both of which focus on teacher retention. Khalid (2025) finds that FDSW adoption is associated with a 3 percentage point increase in teacher retention and a corresponding decrease in new hires. Effects are largest among early-career teachers and two years post-adoption. Similarly, Lawson (2025) finds that FDSW adoption reduces teacher turnover by 2.7 percentage points. Additionally, he finds limited evidence that FDSW adoption affects characteristics of new hires, including experience and education.

2.2 Literature Review

The literature analyzing the impact of four-day school weeks on students, teachers, and broader communities is small but growing. Our paper fits in with a handful of quantitative studies that assess the causal impact of implementing a FDSW. Like our work, most of these studies look at FDSW adoption in one state or a handful of states and implement difference-in-differences methodologies to estimate impacts of this policy. We focus our review on two groups of outcomes that have been studied by other researchers: teacher retention and student achievement.

2.2.1 Teachers and Schools

The literature on FDSWs and teachers is growing as many districts are adopting FDSWs as a teacher retention and recruitment tool. Results on how FDSWs impact teacher retention and recruitment are mixed. Camp (2024), Khalid (2025), and Lawson (2025) find evidence of increased teacher retention after FDSW adoption. Camp (2024) also finds some suggestive positive evidence on teacher quality, while Lawson (2025) reports no effects on experience or education of new hires. Thompson (2025) studies FDSW adoption in Missouri and Oklahoma and finds no overall effect on teacher retention but small positive effects among teachers who would have otherwise moved to a FDSW district. He also finds that greater saturation of FDSW districts in a local area increases teacher turnover, driven by teachers moving from traditional districts to nearby FDSW

districts. Ainsworth et al. (2024) finds increased teacher turnover after FDSW adoption in Oregon, due to a mix of moves out of FDSW districts and exits from the teaching profession altogether.

Work in this area has also estimated the effect of adopting a FDSW on school finances and expenditures. Cost-saving is often mentioned as a rationale for adopting a FDSW, especially in the earliest instances (Thompson, 2021b). Griffith (2011) estimates the highest potential cost savings a district could expect as 5.4%. Most of those savings are through operations/maintenance and school administration, as these districts do not cut educator pay or benefits. Empirically, Thompson (2021a) finds that FDSWs lead to a 1-2% reduction in district expenditures and Morton (2021) finds a reduction in the range of 2% but finds no impacts on per-pupil expenditures. These reductions come through non-instructional support and service expenditures, as well as lower federal funding due to schools paying for fewer free and reduced price lunches which are subsidized by the federal government. While it does appear districts can expect some cost savings from adopting a four day week, the savings are minimal.

2.2.2 Student Achievement

The evidence on the impact of FDSWs on student outcomes is mixed, though the most comprehensive work points to negative or no effects. Only a few studies find a positive relationship between FDSWs and student achievement (Hewitt and Denny, 2011; Anderson and Walker, 2015; Barnes and McKenzie, 2023). Studies in Oregon have found that adopting a FDSW leads to reduced math test scores across elementary, middle, and high school, and a decrease in reading scores in earlier grades, though these effects are noisy (Thompson, 2021b; Thompson et al., 2021; Morton, 2021). Studies that pool across multiple states have found negative achievement impacts in both reading and math test scores (Morton et al., 2024; Thompson and Ward, 2022; Kilburn et al., 2021)⁵. This prior literature has highlighted time in school as a key reason for decreased achievement in FDSWs. A study of FDSWs in 12 states found overall negative effects of FDSW adoption but heterogeneity by time in school. Negative effects were concentrated among the schools that had the shortest school days and effects were small and insignificant for FDSWs that maintained longer time in school (Thompson, 2021b; Thompson and Ward, 2022). Additionally, impacts of FDSWs seem to differ by district type, with negative effects concentrated in non-rural school districts (Morton et al., 2024; Thompson et al., 2022).

2.2.3 Other Student-Level Outcomes

While achievement is the most natural outcome to study in this context, there are many other student-level outcomes that could be impacted by the shift to a FDSW. Low attendance is often cited in the rationale for adopting a FDSW, though there is not evidence of significant attendance effects following FDSW adoption (Morton, 2023; Anderson and Walker, 2015; Morton, 2021; Thompson et al., 2022). In Oklahoma, Morton (2023) finds that adoption of a FDSW led to a 39% reduction in bullying. Outside of school, Fischer and

⁵See Table 1 in Morton et al., 2024 for a summary of these findings.

Argyle (2018) show that FDSWs in Colorado led to an increase in juvenile crime by 20%, driven by property crimes.

Existing work has also studied the impact of FDSWs on student health. In Oregon, Tomayko et al. (2021) found that FDSWs were associated with higher food insecurity among students, higher consumption of sugary beverages, and increased marijuana use, while Tomayko et al. (2020) found no impact of FDSWs on childhood obesity. In Colorado, Israel et al. (2020) finds mixed results; FDSWs are associated with decreased substance use and decreased screen time but also with less sleep, lower breakfast consumption, and higher rates of bullying and sexual activity. While there is not much consensus among these findings, these results underline the potentially broad-reaching effects of FDSWs.

Our study extends this literature on FDSWs to a new context. We are among the first to study FDSWs in Texas, the state with the largest teacher workforce and second largest student population in the country. This paper provides additional evidence on teacher retention and recruitment in Texas, contributing to this existing literature, and provides the first credible estimates of the causal effects of FDSWs on student achievement in Texas. Our conclusions from the teacher retention analysis are consistent with existing work, though our results around teacher recruitment differ, likely due to differences in control group selection. Beyond student achievement and teacher retention, this paper is the first to examine college-going behavior and student work behavior following FDSW adoption. Together, these findings offer the most comprehensive causal evidence to date on the effects of FDSWs in Texas.

3 Data

The data used in this study come from the University of Houston Education Research Center (UH ERC), which hosts linked student-level data from students across the state of Texas. The UH ERC combines data from various sources: K-12 data from the Texas Education Agency (TEA), higher education data from the Texas Higher Education Coordinating Board (THECB) and the National Student Clearinghouse (NSC), and employment data from the Texas Workforce Commission (TWC). The student-level K-12 data include demographics (ethnicity/race, gender, and flags for free or reduced price lunch (FRPL), limited English proficiency (LEP), and English as a second language (ESL) status), enrollment and attendance, discipline, and state standardized tests. The K-12 data also contain information on staff employment, which is our source of data on teachers. These data include a teacher's district and school of employment, demographics, pay, and credentials. The data also allow for student-to-teacher matching, which we use to estimate value-added as a measure of teacher quality. The college-going data contain detailed information about enrollment at Texas colleges and universities, supplemented with less-detailed enrollment data from the National Student Clearinghouse, which can be used to identify out-of-Texas college enrollment. Finally, the employment data come from Texas Unemployment Insurance wage reports. Data is available from these sources on a six month lag for the employment data, and up to two years later for the education data.

3.1 Outcome Definition

3.1.1 Teacher Outcomes

Our teacher sample contains individuals who are identified as either teachers or substitute teachers. To study teacher retention, we follow Ronfeldt et al. (2013) and define three discrete teacher outcomes: stayer, mover, leaver. A stayer is a teacher who is observed in the same district in years t and $t + 1$. A mover is a teacher who is observed in a different district in year t than in year $t + 1$. A leaver is a teacher who left the Texas public school system in year $t + 1$. Our main outcomes of interest are the share of stayers, movers, and leavers in each district.

In addition to retention, we are interested in how teacher quality is affected by FDSW adoption. Outcomes of interest include teacher experience, tenure, education, and certification. We calculate experience as the number of prior years that a teacher is observed in the data. We calculate tenure as the number of prior years that a teacher is observed within a district. Experience and tenure are left-censored at 2006, the earliest year of our data, so values should be interpreted as lower bounds on true years of experience and tenure those teaching before our data begins. We define a new teacher as having zero years of experience and a new hire as having zero years of tenure in the district. For education, we create an indicator for whether a teacher has a Master's degree or higher. For certification, we categorize teachers into three types: standard, emergency, or probationary. Standard teaching certifications are typically obtained through a university degree or an educator preparation program and must be renewed every five years. Emergency certificates are valid for one year and serve to fill a position when a certified teacher cannot be found. Probationary certifications are given to teachers who are working towards the standard teaching certificate (through an educator preparation program) and are valid for one year.

To investigate effects on teacher recruitment, we construct variables for teacher quality among new hires in a district. These variables capture the average characteristics of teachers recruited by districts each year. We also compute a measure of pre-FDSW teacher value-added using 2012-2019 data⁶. To estimate value-added, test scores are residualized using a vector of controls which include two years of lagged scores, student characteristics (race/ethnicity, gender, FRPL, ESL, SPED), and grade-by-year and school fixed effects. We standardize value-added across all teacher-years, so estimates are interpreted as shifts in the teacher quality distribution rather than in student test scores. We take the average 2012-2019 value-added as our measure of pre-FDSW teacher quality, allowing us to investigate whether teachers sort differentially to FDSW districts.

⁶We match students to their math and reading teachers using the class identifier in the class roster data from TEA. To identify each student's math teacher, we restrict to classes with a mathematics subject area. If a student has multiple math classes, we select the class in which the student has most enrolled days by at least 50%. If a student has no math classes, we use self-contained classes, matching students to the teacher with which the student has all or most enrolled days by at least 50%. We follow a similar process to identify each student's reading teacher. We then merge the matched teachers and students with standardized STAAR test scores.

3.1.2 Student Outcomes

Our student outcomes are various measures of academic achievement and other experiences in school, all of which we aggregate to the district level. If a student is enrolled in multiple districts in a given year, we assign students to their most recently attended district. Our primary measure of student achievement is state standardized test scores. Texas began using the State of Texas Assessment of Academic Readiness (STAAR) tests in 2012. End-of-grade tests are administered annually for students in grades 3-8 and are meant to assess grade-level proficiency in various subjects.⁷ We focus on the math and reading components of these standardized tests. For each test, we normalize student test scores within subject, grade, year, and testing language (English or Spanish). If students have multiple scores of the same test in a district-year, we take the earliest score. We then collapse this measure to the district-year level to get an average standardized score. Each year, TEA also determines test score cutoffs to categorize students into four performance levels: did not meet, approaches, meets, and masters grade level. We compute the share of students in a district-year that meet grade-level standards for math and reading as a measure of proficiency. We additionally have end-of-course test scores for English 1 and Algebra 1, which are administered when the course is taken instead of in a specific grade. We compute standardized measures using the same process described above among the sample of high school students who took those exams.

We construct measures of high school graduation rate and college-going behavior using graduation data from TEA and college enrollment data from THECB and NSC. We use a cohort-based approach to identify the graduating class for each academic year based on when they entered 9th grade. For each year, we define a 9th grade cohort as all students enrolled in 9th grade for the first time; their expected graduation is 3 years after (e.g., ninth grade students in AY2019, the 2018-19 academic year, are in the cohort that is expected to graduate in 2022).⁸ For a given district-year, the graduation rate is the share of the cohort that graduated on time, defined as graduating in or before their expected graduation year. To measure college-going behavior, we merge data on college enrollment with data on high school graduates to identify whether a student enrolled in any post-secondary institution (including community colleges, 2-year, and 4-year institutions) in the year following their graduation.

Additional student outcomes we construct are attendance, discipline, mobility, and work behavior. Attendance rate is calculated by TEA as the number of days a student was present out of the number of days a student was in membership (either marked as present or absent). We construct a measure for chronic absenteeism which is defined in Texas as missing 10% or more of school days. Data on discipline includes information on the disciplinary action (suspension, expulsion, etc.) and the reason (violence, substance, conduct, etc.). We construct two measures of discipline for each student, whether a student has any disciplinary action and the number of disciplinary actions, conditional on having any. This allows us to capture

⁷The STAAR test is administered in reading, writing, math, science and social studies.

⁸We add students to the cohort if they transfer in at a later grade with the same expected graduation (e.g., a 10th grade transfer student in AY2020 is added to the 2022 cohort). We drop students who leave the district or state if their exit year is before their expected graduation using the TEA leavers file, which identifies student exits from a district. See [this website](#) for additional details on how a leaver is defined.

both the extensive margin and intensive margin of how a reduced calendar could affect student conduct. To capture student mobility, we compute the share of students who are enrolled in the same district in years t and $t + 1$. We use data from TWC to measure work behavior among high school students. We define a student as having any employment if they appear in the TWC data. Because earnings are reported at the quarterly level, we calculate total earnings for a given academic year as the sum from Q1 and Q2 in the current year and Q3 and Q4 from the previous year (for example, earnings for AY2019 include July-Dec of 2018 and Jan-June of 2019).

3.2 Sample Definition

To identify districts that adopted a four-day school week, we start with a list compiled by Priyanti et al. (2024). We hand checked the list with district websites and calendars to create a preliminary list of FDSW districts and year of adoption. We then used TEA attendance data to identify additional districts with four-day school weeks. We flagged districts in the data that had large drops in the average number of total days taught between 2015 to 2021⁹ We verified websites of districts with large drops and identified additional FDSWs. Table 1 shows the number of districts in Texas that have a FDSW by academic year. We exclude hybrid districts that only have a FDSW for a portion of the academic year, such as during the Fall semester or every other week. The first districts adopted four-day weeks in AY2020. The number of districts with a FDSW has grown quickly; by AY2025, the last year in our data, 176 districts across the state had a FDSW calendar. Figure 1 shows the distribution of FDSW districts across the state by year of adoption.

The first set of columns of Table 2 present district, student and teacher-level characteristics from AY2019 for districts that never adopted a FDSW and districts that eventually adopted a FDSW. The first column presents the mean of the characteristic for traditional (five-day) districts and the second presents the mean for four-day districts. The third column is the difference between four-day and five-day districts. Consistent with other states, FDSW districts are largely in rural areas. Nearly 80% of districts that adopted a FDSW in Texas are in rural areas, compared to 59% of traditional districts. FDSW districts are smaller than traditional districts, as measured by both the number of students enrolled (947 versus 5,876) and the number of schools in the district (2.5 versus 8.4). FDSW districts had fewer instructional days for students even before making the switch to a four-day week, with 162.8 compared to 165.0 days taught, on average.¹⁰

FDSW districts employed a higher proportion of white teachers (82% versus 73%) and a lower proportion of teachers with standard certification (88% versus 90%), meaning more teachers have an emergency or probationary certification. Additionally, FDSW districts had a lower share of teachers with a Master’s degree (15% versus 17%) and lower average salaries by \$2,495 per year. Teachers in FDSW districts had approximately 0.5 years less tenure in their district and slightly less overall teaching experience (0.2 years),

⁹TEA documentation defines “total days taught” as the number of days of instruction offered to each student at a school as shown on the school calendar. This corresponds to the number of instructional days in a district with the exception of students who transfer into or out of a district.

¹⁰This measure is constructed by taking the district-level average of each student’s total days receiving instruction, so it is a noisy but informative measure of the number of school days in an academic year.

consistent with the higher share of new hires and new teachers.

Student populations between traditional and FDSW districts also look different. FDSW districts had a higher proportion of white students (57% versus 48%) and a lower share of students classified as having Limited English Proficiency (10% versus 14%). They also have a higher share of students receiving free or reduced-price lunch (58% versus 53%). Students in FDSW districts performed worse on state standardized tests in math and reading. They were 3.2 percentage points less likely to graduate high school and 4.5 percentage points less likely to attend college.

These comparisons highlight systematic differences between traditional and FDSW districts, with FDSW districts being smaller, having greater teacher turnover, and performing worse academically than districts that do not adopt a FDSW. This emphasizes the importance of choosing control districts that provide a plausible counterfactual to FDSW districts. In the next section, we discuss our methodology for constructing a control group through propensity score matching.

4 Empirical Strategy

We use a staggered difference-in-differences (DiD) design to estimate the causal effects of FDSW adoption on student and teacher outcomes. A staggered DiD leverages differences in adoption of a FDSW across districts and across time. Our design compares outcomes at FDSW districts, before and after adopting the policy, to districts that did not adopt a FDSW. This allows us to account for time-invariant district characteristics, as well as common trends across districts.

4.1 Control Group Selection

As seen in the first set of columns of Table 2, districts that adopt a FDSW are observably different from those that do not; FDSW districts tend to be small and rural, and have higher teacher turnover and lower student achievement than traditional five-day districts. To ensure that we are comparing FDSW districts to traditional districts with similar trends, we build a matched control group using nearest neighbor propensity score matching. We estimate the propensity to adopt a FDSW as follows:

$$Pr(\text{FDSW}_d = 1 | X_{d,2012}, X_{d,2019}, \Delta X_{dy}) = \Lambda(\alpha + \beta'_{2012} X_{d,2012} + \beta'_{2019} X_{d,2019} + \sum_{y=2013}^{2019} \gamma'_y \Delta X_{dy})$$

where FDSW_d is an indicator for whether the district ever adopted a four-day school week, Λ is the logistic link function, $X_{d,y}$ is a vector of district, teacher, and student characteristics, and student outcomes from Table 2,¹¹ and $\Delta X_{d,y}$ are year-to-year differences from 2012 to 2019 in STAAR math and reading scores and

¹¹This vector includes variables for district locale, size, and instructional days; student gender, race and ethnicity, free and reduced price lunch status, and Limited English Proficiency status; teacher gender, race and ethnicity, retention, experience, tenure, pay, certification, and education; STAAR math and reading scores, graduation rate, and college enrollment. To minimize overfitting, we prioritize variables with a statistically significant difference between traditional and FDSW districts and exclude variables that are rare.

teacher retention to capture trends in these outcomes (e.g., $\text{STAAR math}_{d,2013}$ is the difference between average STAAR math score in district d in 2013 and average STAAR math score in district d in 2012). The coefficient vectors are estimated using logistic LASSO, which selects a subset of covariates based on their predictive power for FDSW adoption.¹² We then use the model selected by the logistic LASSO to estimate a propensity score of FDSW adoption for both treated and control districts, which can be interpreted as the likelihood of adopting a FDSW given the 2012-2019 values and trends for a given district.

To build our control group, we match each FDSW district to its two nearest neighbors (traditional districts with the closest propensity scores), given that the propensity score is within 0.01 of the FDSW district's score. This process leads to the removal of 6 treated districts which did not have a close enough control district, while the rest of the districts match to 230 control districts. The second set of columns of Table 2 presents district, teacher, and student-level characteristics, and student outcomes for our sample of matched control and treated districts. The set of control districts are observably similar to treated districts, with the exception of marginally significant differences in number of school and share of Black teachers. The overall balance across observable characteristics in AY2019, though not required for our empirical design, supports the assumption that matched control districts serve as a plausible counterfactual for what would have happened if treated districts had not adopted a FDSW.

4.2 Methods

We implement our staggered DiD model using the Callaway and Sant'Anna (2021) method. In this method, each treated district d is assigned to a group g which corresponds to treatment timing, so all districts that adopted a FDSW in the same academic year are in the same group. The Average Treatment-on-the-Treated (ATT) effects are then estimated for each group-by-time (g, t) combination. The overall ATT is calculated as an average of all post-treatment ($r \geq 0$, where $r = t - g$ is the year relative to adoption) estimates, weighted by the group size.

Identification of causal effects requires the standard staggered difference-in-differences assumptions, including the parallel trends assumption. Parallel trends in this setting requires that in the absence of adopting a FDSW, outcomes for FDSW districts would have trended similarly as outcomes for the districts that did not adopt a FDSW. Our primary evidence in support of this assumption is shown in the event-study estimates from the pre-period ($r < 0$), or the periods before FDSWs were adopted. These estimates test whether, on average, the difference in pre-trends between treated and control districts are indistinguishable from zero. These tests, in conjunction with the balance we achieved in the control group construction, allow us to assess whether the parallel trends assumption is plausible.

¹²The selected variables include (1) 2012 values for NCES rural, suburb, and city indicators, share of female, Hispanic, and Asian students, share of students with FRPL status, share of Hispanic teachers, share of movers, average 3rd grade STAAR math and reading scores, average 5th grade reading score; (2) 2019 values for number of instructional days, NCES city indicator, number of elementary schools, share of white and Hispanic students, share of students with LEP status, share of Hispanic teachers, share of movers, average teacher tenure and pay, share of teachers with standard certification, average 3rd grade STAAR math score; (3) year-over-year differences in stayers, movers, and STAAR math and reading scores (overall, 3rd grade, and 5th grade).

An additional key assumption is no anticipation. We believe this assumption is valid in terms of our student outcomes, as it is unlikely that test scores or behavioral outcomes would change before students have the extra day off of school. There is potential for anticipation in our teacher outcomes. Because districts typically announce four-day school week policies the year before adoption, teachers may move districts in anticipation of the change. We see some evidence of anticipatory effects in the year before adoption ($g - 1$) for teacher retention outcomes. To account for this, we present results with treatment shifted back one year to align with the time of announcement rather than adoption.

5 Results

5.1 Validation

We first present results in support of the parallel trends assumption underlying our staggered DiD design, which assumes that outcomes in FDSW districts would have had similar trends as in control districts if they had not adopted a four-day school week. We plot the pre-period event study estimates and show that the estimates are generally statistically indistinguishable from zero. These estimates show that outcomes in treated and control districts were trending similarly prior to FDSW adoption, consistent with the assumption that treated districts would have had similar trends to control districts had they not adopted a FDSW. This pattern holds for both student and teacher outcomes, supporting the interpretation that post-treatment differences in outcomes can be credibly interpreted as the causal impact of FDSW adoption rather than due to underlying differences between FDSW and traditional districts.

We next verify that FDSW adoption did in fact reduce the number of instructional days in treated districts. Figure 3 presents an event study of the average days taught in a school year. Prior to FDSW adoption, the point estimates are generally statistically indistinguishable from zero, in line with our parallel trends assumption. Upon adoption, we observe a sharp and statistically significant decline of approximately 13.5 days. FDSW adoption significantly reduces the number of instructional days, albeit by less than the number of weeks in a typical school year. Districts across the state were reducing the number of instructional days in their calendars once the passage of HB 2610 removed the minimum day requirement. Because of this, our estimate captures the reduction in instructional days from FDSW adoption relative to districts that were also reducing days without moving to a four-day week.

5.2 Teacher Retention

Worsening teacher turnover was an explicit motivation of many districts adopting FDSWs in Texas. Our main teacher outcomes are the three measures of retention defined previously: stayer, mover, and leaver. As previously discussed, we adjust for anticipatory effects due to the timing of the policy announcement by shifting treatment back one year for these teacher outcomes. We find evidence that FDSW adoption has a

significant and positive impact on retention. Figure 4 presents the event study for the share of teachers who stay in their district in the following year. FDSW adoption increases the share of stayers by 2.1 percentage points, or 2.6% from the control mean. These results are driven by a -1.7 percentage point change in the share of teachers who move to another district (16.7% decrease from the control mean); we find a small and insignificant decrease of 0.5 percentage points on the share of teachers who leave the profession. Figures 5 and 6 present the event studies for movers and leavers, respectively. These results suggest that FDSW adoption can increase teacher retention by making the district more attractive than competing districts, but we do not find strong evidence that FDSW adoption retains teachers in the profession.

The magnitude of our retention estimate is generally consistent with the literature linking teacher pay to retention. As shown in Table 3, we do not find changes to teacher salaries following FDSW adoption. However, the reduction in instructional days can be thought of as an increase in effective daily wages. Using an average teacher salary among control districts of \$41,668 and a reduction in days taught by 13.5 from the control mean of 163 days, this translates to an effective increase in daily compensation of roughly 9%. Sun et al. (2025) finds that district-wide salary increases in Washington ranging from 15% to 20% raised retention by 1 to 3 percentage points. Targeted retention bonuses in Tennessee and North Carolina that were approximately 6% to 15% of teacher salaries produced much larger effects, ranging from 5 to 23 percentage points (Clotfelter et al., 2008; Springer et al., 2016; Swain et al., 2019). Because a four-day week applies to all teachers in a district, we view it as more comparable to a district-wide salary increase. Our findings suggest that four-day school weeks can be at least as effective as district-wide salary increases at improving teacher retention without requiring increases in district spending.

5.2.1 Teacher Subgroups

We investigate retention effects among various subgroups of teachers who may respond differently to aspects of the FDSW. These results are presented in the first panel of Table 3. First, we investigate teachers by bins of experience: less than 5 years, 5 to 10 years, and more than 10 years. Early-career teachers tend to have higher turnover (Redding and Henry, 2018). As they are learning the profession, these teachers may benefit more from the additional planning time that a FDSW provides. We estimate larger effects among early- and mid-career teachers (2.2 and 2.4 percentage points, respectively) compared to late-career teachers (1.6 percentage points). Though the estimates are not statistically distinguishable from each other, they are consistent with less experienced teachers valuing the reduced instructional time, which could allow for more planning.

Additionally, we consider the effects of a FDSW on teachers who are also parents to young children. The flexibility offered by a FDSW may be more valued by these parents. Parental status of teachers is not reported in the data, so we roughly approximate this using teacher age. We define a subgroup of teachers who are more likely to be parents of young children as those who are 25-40 (“parent age”) and compare them to teachers over 40 (“non-parent age”). Parent-age teachers are 2.7 percentage points more likely to

stay if their district adopts a FDSW, compared to 1.1 percentage points among non-parent age teachers. Again, these results are not statistically distinguishable from each other but are consistent with parents of young children valuing the flexibility offered by a FDSW.

5.2.2 Robustness: Spillovers to Neighboring Districts

One potential concern when estimating teacher retention effects is spillovers across treated and control districts. As shown in Figure 2, many of the matched control districts are neighboring a FDSW district. If teachers move from nearby control districts to FDSW districts following adoption, retention in control districts may decline. This would inflate estimated retention gains for treated districts and represent a violation of the stable unit treatment value assumption (SUTVA) necessary for the staggered difference-in-differences design to be valid. To investigate whether this is an issue in our setting, we conduct a robustness check using a control group of districts that are never neighboring a FDSW district. These control districts do not share a border with a district that adopted a FDSW within our sample time frame, and thus, are less likely to be affected by spillovers. The results from this check are presented in Appendix B and show that our main results are consistent across these different definitions of the control group. The effects on retention are nearly identical when using the no-neighbor control group as when using our preferred matched control group, alleviating concerns around potential spillovers.

5.3 Teacher Recruitment

Recruiting higher quality teachers was part of the motivation for adopting a FDSW. We find that FDSW adoption leads to a 1.1 percentage point reduction in the share of teachers in a district who are new hires. This is consistent with these districts having fewer vacancies due to increased retention among incumbent teachers. The composition of new hires at FDSW districts could change either because fewer vacancies allow districts to be more selective or by expanding their applicant pool, if teachers are looking to move to a FDSW district. We look at characteristics of new hires as outcomes, presented in Panel B of Table 3. We find that newly hired teachers at FDSW districts have more experience and higher educational attainment. New hires are 6.1 percentage points less likely to be first-year teachers, have an additional half year of experience, and are 2.8 percentage points more likely to hold a Master’s degree. These results suggest that districts are hiring observably higher quality teachers.

To more directly investigate whether new hires are more effective, we look at value-added scores. Our pre-FDSW value-added measure allows us to see if districts are recruiting better teachers based on their prior performance. This measure is not contaminated by contemporaneous effects that FDSWs can have on teacher effectiveness or student achievement.¹³ We thus are only able to estimate pre-FDSW value-added

¹³FDSWs may also lead to improved effectiveness of existing teachers (due to increasing time for lesson planning or professional development). We are unable to directly estimate dynamic effects on teacher value-added. Our identification strategy leverages district-level variation in the adoption of four-day school weeks, making it impossible to distinguish changes in student test scores due to teacher improvement from other effects of four-day school weeks.

among new hires who taught math or reading in Texas between 2012 and 2019. We find evidence that FDSW districts hire more effective math teachers (0.378 SD) but reading teachers are no more effective. While we cannot measure the direct impacts of FDSW adoption on teacher effectiveness, our findings are consistent with districts recruiting higher quality teachers, both in terms of observable characteristics and effectiveness.

5.4 Student Achievement

Our main student outcome is pooled standardized test scores from 3rd to 8th grade, averaged at the district level. Due to the COVID-19 pandemic, the STAAR test was not administered in AY2020. We shift testing data after 2020 back one year, so relative time is the number of *testing* years since adoption.¹⁴ Overall, we do not find significant effects on student achievement, across grade levels and subject. We present our main effects on pooled STAAR math scores in Figures 7 and 8. Figure 7 presents impacts on standardized STAAR math scores and Figure 8 presents impacts on the share of students who score above math standards on the exam. We do not find evidence of impacts on either of these measures of math achievement up to three years after the adoption of a FDSW. Figures 9 and 10 present results for reading scores. The overall ATT on reading scores is small and insignificant (-0.006), but the event study in Figure 9 shows that there is a positive effect on reading achievement in the final relative period in our data, though this is likely driven by early-adopting districts. We explore this in greater detail in Appendix A. In Panel A of Table 4, we break out test score results separately for elementary school and middle school grades, as well as for Algebra 1 and English 1 in high school. As with the pooled results, we do not find significant effects on test scores within these grade levels. Though we find null results using the overall sample, these results disguise heterogeneity across districts, which we investigate in Sections 5.4.1 and 5.4.2.

The effect of a FDSW on student achievement could be mediated by various factors. For instance, FDSW adoption could lead to changes in student composition due to families moving into or out of the district. We do not find significant effects on student mobility out of the district, presented in Panel B of Table 4. We also estimate the effect of FDSW adoption on the student-to-teacher ratio. Given that FDSW districts are able to improve teacher retention, there could be a subsequent reduction in class sizes. We find insignificant effects. It appears that neither student sorting across districts of reduced class sizes are contributing to the impacts of FDSWs on student achievement.

In Figures 11 and 12, we compare our estimates for effects on math and reading scores, respectively, against effects from the FDSW literature from other states. Our main pooled result on either math scores or reading scores is presented in the center of the plot, in blue. The rest of the plot presents estimates from the literature, noting the paper each estimate comes from and the state studied. All of these results are from standardized exams; the Morton (2021), Thompson (2021b), and Thompson and Ward (2022) results are

¹⁴In AY2021, Texas resumed in-person testing, but students that were still remote could essentially opt out of the test by staying home (see [this](#) article). Districts in our sample had, on average, fewer remote days and higher participation in testing as compared to the rest of the districts across the state. We observe that these districts also had higher test scores, which is consistent with in-person versus remote learning (as found in [this](#) report), though trends are similar between the districts in our treatment and control groups.

pooled across grades 3-8, while Morton et al. (2024) is pooled K-8 scores. Our estimates for math scores are consistent with prior work on four-day school weeks, most of which finds small negative effects that are statistically insignificant at conventional levels. Our point estimate, however, is closer to zero and more precisely estimated than those in the existing literature. For reading, our results diverge more clearly from prior findings: most studies report negative effects, whereas we find a precisely estimated null effect. While concerns about the impacts of FDSWs on student learning are warranted, our results indicate that negative effects on achievement are not inevitable. As we show in the heterogeneity analysis that follows, the effects of FDSWs on student outcomes depend on the context in which the policy is adopted and its implementation.

5.4.1 Heterogeneity: By Instructional Time

In this section, we explore heterogeneity in the effects of FDSWs on student outcomes based on total time in school after adoption. The most direct mechanism for FDSWs to impact student outcomes is through reduced instructional time. Whether the move to a FDSW results in a significant decrease in instructional time depends on how districts adjust the length of the remaining school days. If districts maintain the same amount of instructional minutes even with the shorter school week, students outcomes may not be as negatively affected. However, if FDSW adoption results in a substantial decline in instructional time, we may expect negative effects on student outcomes. Consistent with this hypothesis, the literature on four-day school weeks attributes negative achievement effects to the loss of instructional time. Thompson (2021b) and Thompson and Ward (2022) find that negative effects are concentrated among districts that reduce instructional time, rather than districts that extend the remaining four days to preserve overall instructional minutes. More generally, existing estimates from reduced instructional days are consistent with worsening student outcomes (Marcotte & Hemelt, 2008).

To investigate whether instructional time drives negative effects of FDSW adoption in our setting, we split our sample of FDSW districts into subgroups by monthly minutes. Data on monthly minutes comes from TEA and are defined as the number of minutes devoted to a particular class. To proxy for instructional time, we restrict to full-time teachers and calculate the total number of monthly minutes across their classes in a year.¹⁵ We then take the average of total monthly minutes in a district. We categorize a FDSW district as having Low Minutes if they are in the bottom tercile of monthly minutes in 2024 and the remaining FDSW districts as Other.

Table 5 presents baseline characteristics of Low Minutes districts compared with Other districts using AY2019 data. There are 49 districts in the Low Minutes sample, compared to 97 in the Other sample.¹⁶ Low Minutes districts are similar to Other districts across district, teacher, and student characteristics. We re-estimate our main specifications separately for each subgroup. The control group is the same set of 230 matched districts discussed in Section 4.1. Low Minutes FDSW districts decreased instructional days by 12

¹⁵This measure is the number of minutes that a teacher is assigned to a service and does not reflect the overall instructional time for students, so we interpret this as a proxy for instructional time.

¹⁶24 districts are missing data on monthly minutes.

days, while Other FDSW districts had 14 fewer days. However, only Low Minute FDSW districts experienced a significant change in instructional minutes; these districts saw a decline of 1,625 monthly minutes, or a 19.9% decrease from the control mean.

Across our main student outcomes, we find a substantial heterogeneity based on instructional minutes. Figures 17 and 18 show the event study plots for math achievement and Figures 19 and 20 show the event study plots for reading achievement.¹⁷ Pre-treatment coefficients for these outcomes are generally close to zero and stable in the years before adoption, which supports the parallel trends assumption within these subgroups. Among Low Minutes districts, we find significant negative effects in math and reading achievement. Math scores decline by 0.038 standard deviations and proficiency falls by 1.8 percentage points. Reading scores decline by 0.047 standard deviations and proficiency falls by 3.2 percentage points. In contrast, districts that preserved more instructional time (Other) show no evidence of the achievement declines, and in some cases, effects even trend positive. Effects on math scores and proficiency are positive but not significant among Other districts. While the effect on reading scores is not significant, there is a marginally significant increase of 1.3 percentage points on proficiency.

Table 6 reports results by grade level. Among Low Minutes districts, FDSW adoption is associated with significant negative effects on middle school achievement, including a decline of 0.069 standard deviations in math scores and 3.0 percentage points in math proficiency, as well as a decline of 0.078 standard deviations in reading scores and 4.3 percentage points in reading proficiency. These effects are nearly twice as large as the pooled estimates presented in our main analysis, suggesting that the modest aggregate effects we find mask heterogeneity in how FDSWs affect older students. Other districts continue to show no evidence of negative effects in either elementary or middle grades. However, elementary reading proficiency and middle school math proficiency actually increase significantly among these districts, with a gain of 1.8 percentage points and 2.1 percentage points, respectively.

Taken together, these results provide suggestive evidence that the amount of instructional time under a four-day calendar is an important moderator of FDSW effects on student achievement in Texas, consistent with findings from the broader FDSW literature. Districts that reduce instructional time the most appear to experience the largest academic losses, particularly in the middle school, while districts that extend school days enough to offset the lost day show small and, in some cases, even positive achievement effects. This heterogeneity suggests that the design of a four-day calendar has meaningful consequences for students.

5.4.2 Heterogeneity: By Accountability Score

In this section, we explore heterogeneity by district quality. We envision two possible scenarios under which a district could decide to adopt a FDSW: one in which a district is specifically experiencing issues with teacher turnover, and another in which teacher turnover is potentially a consequence of other challenges at

¹⁷We estimate coefficients for all periods but do not plot the estimates for $r = 3$ due to too few treated districts.

the district, such as funding constraints. If districts were struggling with more than just teacher turnover, adopting a four-day school week may exacerbate underlying problems even if teacher retention is improved. In this case, we would expect FDSW adoption to more negatively affect students, whereas using this policy as a targeted response to turnover issues could have smaller effects on student performance. To proxy for district quality, we create a “struggling district” measure using state constructed accountability scores.

Accountability scores are calculated by the TEA and could reflect the the broader challenges of a district despite not directly measuring them. Districts with strong accountability ratings prior to adopting a FDSW were plausibly targeting teacher retention and recruitment challenges. In this setting, we might expect a FDSW to function as a amenity for teachers, with limited spillover to student outcomes. On the other hand, districts facing a wider set of challenges beyond teacher turnover may potentially compound existing struggles by adopting a four-day school week. To examine the role of district quality, we split our sample of FDSW districts by their 2019 TEA overall accountability score. The accountability score measures district performance in three domains: (1) Student Achievement, (2) School Progress, and (3) Closing the Gaps. Student Achievement includes STAAR assessments, college, career, and military readiness, and graduation rates. School Progress includes student growth on STAAR assessments and relative performance compared to similarly economically disadvantaged campuses. Closing the Gaps tracks differentials among demographic and socioeconomic groups. We define districts in the bottom quartile of the overall score as Struggling districts and all other FDSW districts as Other.

Table 7 presents AY2019 district, teacher, and student characteristics, and student outcomes, for these two subgroups prior to FDSW adoption. There are 35 Struggling districts and 135 Other districts. Importantly, the Struggling district subgroup is distinct from the Low Minutes subgroup, with only 13 FDSW districts in both. Struggling districts serve more diverse student populations, have a higher share of students eligible for free or reduced-price lunch, and have higher rates of chronic absenteeism prior to adoption. Struggling districts also have less experienced teachers, with lower average tenure, a lower share of teachers with standard certification, and worse retention. As expected, Struggling districts have worse baseline student achievement across various measures: overall math and reading scores are roughly a quarter of a standard deviation lower, the shares of students meeting math and reading standards are 12.5 and 9.8 percentage points lower, respectively, and Algebra 1 and English 1 scores, along with college enrollment, are significantly lower than in Other districts. These baseline characteristics provide suggestive evidence that Struggling districts adopted FDSWs in different contexts than Other districts and motivate estimating the effects of adoption separately for each subgroup.

We first examine overall math and reading scores and proficiency measures. Figures 21 and 22 present event study plots for math outcomes and Figures 23 and 24 for reading outcomes.¹⁸ Pre-treatment coefficients are close to zero and statistically indistinguishable from zero in all years before adoption, consistent with the parallel trends assumption for these subgroups. While we do not find significant effects

¹⁸We estimate coefficients for all periods but do not plot the estimates for $r = 3$ due to too few treated districts.

on math outcomes for either group, we find negative effects on reading among Struggling districts, though these estimates are noisy due to the smaller number of districts in this subgroup. Effects on reading scores are not significant on average (-0.025) but the event study plots reveal a declining pattern over time. Reading proficiency declines significantly by 2.8 percentage points. In contrast, Other FDSW districts show no evidence of negative achievement effects. The grade-level results in Table 8 show similar patterns. Struggling districts experience significant declines in reading proficiency for both elementary and middle school.

Taken together, these results support the interpretation that district context at the time of adoption shapes the impact of FDSWs on students. Districts that were already struggling academically and facing broader challenges prior to adoption appear less equipped to compensate for the loss of instructional time. Districts with stronger baseline conditions, plausibly adopting FDSWs as a targeted tool for teacher turnover, show no evidence of negative impacts on students and in several cases realize modest gains in student outcomes. This heterogeneity underscores that a district's conditions are important predictors of whether a four-day school week can effectively improve teacher retention without negatively affecting students.

5.5 Student Behavior and Attainment

Beyond standardized test scores, we examine effects of FDSW adoption on other measures of student engagement and success. Panel B of Table 4 presents aggregate treatment effects for these outcomes. Pre-treatment averages for these outcomes are close to zero and statistically insignificant, consistent with parallel pre-trends. We find no significant effects of FDSW adoption on chronic absenteeism or disciplinary actions. These results on student attendance are consistent with previous literature on FDSWs. Our disciplinary results differ from other work in this literature which finds reductions in disciplinary actions. If reductions in disciplinary incidents are driven by students interacting less in school, we may expect these null effects in Texas given that FDSW districts extended their school days by more than in other states.

We observe a dynamic pattern for student work behavior, high school graduation, and first-year college enrollment. Figure 13 presents effects on the share of high school students with any employment, and Figure 14 presents effects on average annual earnings among high school students. Neither outcome shows a significant aggregate effect (0.006 for employment and $\$96.40$ for earnings), but estimates for the share of students working increase over time. This is consistent with the idea that an additional day off from school may create greater opportunity for part-time work. However, later relative-time estimates are supported by fewer, earlier-adopting cohorts, so we cannot rule out that these patterns are specific to the earlier cohorts. Similarly, for graduation and college enrollment, we find null aggregate effects, but estimates are growing in magnitude and eventually statistically significant in later relative-time periods.

6 Discussion

This paper studies a fundamental shift in the delivery of education, the adoption of a four-day school week. This policy has potentially wide-ranging effects due to the numerous channels through which it affects both teachers and students. Negative effects on student performance could arise as teachers and students are spending one less day together in the classroom; however, teachers may value the extra day of flexibility for tasks such as lesson planning and grading, which could improve teacher effectiveness or attract higher-quality teachers. In light of these various mechanisms, our study is the first to provide a comprehensive picture of the effects of this growing policy in Texas by examining students and teachers simultaneously. Furthermore, the richness of the data allows us to explore effects of FDSWs on student outcomes that have not previously been studied, such as college enrollment and work behavior.

Our main results show that FDSW adoption leads to a meaningful increase in teacher retention: the share of teachers who stay in their district rises by 2.1 percentage points, driven by a substantial decline in the share of teachers moving to other districts. This effect is comparable in magnitude to what districts might expect from a sizable salary increase, suggesting that a four-day schedule functions as a real, non-monetary amenity for teachers. We also find that FDSW districts are able to recruit higher quality new hires. We estimate positive effects on average experience, share with a Master's degree, and prior math value-added of new hires. Our work thus finds impacts consistent with the goal of many districts adopting four-day school weeks: to retain and recruit better teachers.

For students, we find little evidence that FDSW adoption negatively affects achievement: pooled estimates on math and reading scores are small and statistically insignificant. The pooled null effects on achievement, however, masks substantial heterogeneity across districts. We find that the amount of instructional time preserved under a FDSW is an important moderator of achievement effects. Districts that cut the most instructional minutes see significant declines in both math and reading scores, while districts that extend their remaining four days by more do not experience significant changes in student test scores. This finding is consistent with prior work in other states linking negative achievement effects to the loss of time in school rather than to the shortened week itself. We also find that the conditions of a district prior to adopting a FDSW meaningfully shape its effects on students. Districts that were already struggling, as measured by lower pre-adoption accountability ratings, experience declines in reading achievement. In contrast, districts with stronger baseline ratings show no evidence of harm to students and even see improvements in reading outcomes. Together, these two heterogeneity results suggest that FDSWs are most likely to benefit students when districts both preserve instructional time and are not simultaneously contending with broader challenges.

Beyond test scores, we provide novel evidence on student behavior and educational attainment. We find insignificant effects on student absenteeism and disciplinary actions. Although the aggregate effects on high school employment and earnings are small and insignificant, the share of high school students who work rises in the years following adoption. This is consistent with the additional weekday creating

an opportunity for part-time work. Similarly, while the aggregate effect on high school graduation and college enrollment is null, event-study estimates grow in magnitude and become statistically significant in later years, which may reflect the cumulative influence of improved teacher retention and recruitment on students' medium-run outcomes.

As adoption of FDSWs continues to expand across the country, districts considering this policy must carefully weigh the evidence. Our findings suggest that FDSWs are not a universal solution, nor a universal risk: their effects on students depend heavily on implementation choices and the conditions in which the policy is adopted. A district that preserves instructional time by lengthening its remaining four days, and that is not struggling academically, can plausibly use a four-day school week to improve teacher retention with minimal harm to students. A district that reduces instructional time significantly, or that adopts the policy amidst broader issues, risks worsening student achievement. As districts decide whether to adopt a four-day school week, they should be mindful of when and how to implement the policy to support both students and teachers.

References

- Ainsworth, A. J., Penner, E. K., & Liu, Y. (2024). *Less is more: The causal effect of four-day school weeks on employee turnover* (EdWorkingPaper No. 24-1035). Annenberg Institute at Brown University.
- Anderson, D. M., & Walker, M. B. (2015). Does shortening the school week impact student performance? Evidence from the four-day school week. *Education Finance and Policy*, 10(3), 314–349.
- Barnes, K., & McKenzie, S. (2023). *Exploring academic outcomes in Arkansas schools: A study of four-day school week and year-round calendar districts* (Arkansas Education Reports Vol. 20 No. 8). University of Arkansas, Fayetteville. <https://scholarworks.uark.edu/oepreport/103>
- Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200–230.
- Camp, A. (2024). *Teacher retention and quality in the four-day school week* (EdWorkingPaper No. 24-971). Annenberg Institute at Brown University.
- Clotfelter, C. T., Glennie, E., Ladd, H. F., & Vigdor, J. L. (2008). Would higher salaries keep teachers in high-poverty schools? Evidence from a policy intervention in North Carolina. *Journal of Public Economics*, 92(5–6), 1352–1370.
- Fischer, S., & Argyle, D. (2018). Juvenile crime and the four-day school week. *Economics of Education Review*, 64, 31–39.
- Griffith, M. (2011). *What savings are produced by moving to a four-day school week?* (Policy Report No. NJ3). Education Commission of the States. <https://www.ecs.org/clearinghouse/93/69/9369.pdf>
- Hewitt, P. M., & Denny, G. S. (2011). The four-day school week: Impact on student academic performance. *Rural Educator*, 32(2), 23–31.
- Hollingsworth, H. (2023). More school districts adopt 4-day weeks, citing lower costs and better teacher recruitment. <https://www.pbs.org/newshour/education/more-school-districts-adopt-4-day-weeks-citing-lower-costs-and-better-teacher-recruitment>
- Israel, W., Mulitauopele, C., Ma, M., Levinson, A. H., Cikara, L., & Brooks-Russell, A. (2020). Adolescent health behaviors in schools with 4-versus 5-day school weeks. *Journal of School Health*, 90(10), 794–801.
- Khalid, A. (2025). *Non-monetary incentives in education: Investigating the impact of four-day school week on teacher retention and hiring* (Working Paper No. 2025-09). University of Arkansas Department of Education Reform.
- Kilburn, M. R., Phillips, A., Gomez, C. J., Mariano, L. T., Doss, C. J., Troxel, W. M., Morton, E., & Estes, K. (2021). *Does four equal five?: Implementation and outcomes of the four-day school week*. RAND Corporation. https://www.rand.org/pubs/research_reports/RRA373-1.html
- Lawson, M. C. (2025). *The effect of four-day school week adoption on teacher retention and sorting* (EdWorkingPaper No. 25-1372). Annenberg Institute at Brown University.

- Marcotte, D. E., & Hemelt, S. W. (2008). Unscheduled school closings and student performance. *Education Finance and Policy*, 3(3), 316–338.
- Morton, E. (2021). Effects of four-day school weeks on school finance and achievement: Evidence from Oklahoma. *Educational Researcher*, 50(1), 30–40.
- Morton, E. (2023). Effects of 4-day school weeks on older adolescents: Examining impacts of the schedule on academic achievement, attendance, and behavior in high school. *Educational Evaluation and Policy Analysis*, 45(1), 52–78.
- Morton, E., Thompson, P. N., & Kuhfeld, M. (2024). A multi-state, student-level analysis of the effects of the four-day school week on student achievement and growth. *Economics of Education Review*, 100, 102524.
- Priyanti, N., Miraj, F., & Templeton, T. (2024). *Exploring four-day school week adoption in Texas: A thematic analysis* (Working Paper No. #102-24). University of Houston Education Research Center.
- Redding, C., & Henry, G. T. (2018). New evidence on the frequency of teacher turnover: Accounting for within-year turnover. *Educational Researcher*, 47(9), 577–593. <https://doi.org/10.3102/0013189X18814450>
- Ronfeldt, M., Loeb, S., & Wyckoff, J. (2013). How teacher turnover harms student achievement. *American Educational Research Journal*, 50(1), 4–36.
- Springer, M. G., Swain, W. A., & Rodriguez, L. A. (2016). Effective teacher retention bonuses: Evidence from Tennessee. *Educational Evaluation and Policy Analysis*, 38(2), 347–368.
- Sun, M., Candelaria, C. A., Knight, D., LeClair, Z., Kabourek, S. E., & Chang, K. (2025). The effects and local implementation of school finance reforms on teacher salary, hiring, and turnover. *Educational Evaluation and Policy Analysis*, 47(2), 380–412.
- Swain, W. A., Rodriguez, L. A., & Springer, M. G. (2019). Selective retention bonuses for highly effective teachers in high poverty schools: Evidence from Tennessee. *Economics of Education Review*, 68, 148–160.
- Texas Education Agency. (2025). *Texas four-day school week campus analysis, 2022–23 school year*. <https://tea.texas.gov/reports-and-data/program-evaluations/program-evaluations-research-reports/4dsw2022-23.pdf>
- Thompson, P. N. (2021a). Does a day lost equal dollars saved? The effects of four-day school weeks on school district expenditures. *National Tax Journal*, 74(1), 147–183.
- Thompson, P. N. (2021b). Is four less than five? Effects of four-day school weeks on student achievement in Oregon. *Journal of Public Economics*, 193, 104308.
- Thompson, P. N. (2025). Four-day school weeks and teacher mobility: Evidence from Missouri and Oklahoma. *Economics of Education Review*, 109, 102708.
- Thompson, P. N., Gunter, K., Schuna, J. M., & Tomayko, E. J. (2021). Are all four-day school weeks created equal? A national assessment of four-day school week policy adoption and implementation. *Education Finance and Policy*, 16(4), 558–583.

- Thompson, P. N., Tomayko, E. J., Gunter, K. B., & Schuna, J. J. (2022). Impacts of the four-day school week on high school achievement and educational engagement. *Education Economics*, 30(5), 527–539.
- Thompson, P. N., & Ward, J. (2022). Only a matter of time? The role of time in school on four-day school week achievement impacts. *Economics of Education Review*, 86, 102198.
- Tomayko, E. J., Gunter, K. B., Schuna, J. M., & Thompson, P. N. (2020). Effects of four-day school weeks on physical education exposure and childhood obesity. *Journal of Physical Activity and Health*, 17(9), 902–906.
- Tomayko, E. J., Thompson, P. N., Smith, M. C., Gunter, K. B., & Schuna, J. M. J. (2021). Impact of reduced school exposure on adolescent health behaviors and food security: Evidence from 4-day school weeks. *Journal of School Health*, 91(12), 1055–1063.

Tables

Table 1: FDSW Districts by Academic Year

(1)	(2)	(3)
Academic Year	Number of FDSWs	Our Sample
2015	0	0
2016	0	0
2017	0	0
2018	0	0
2019	0	0
2020	4	4
2021	14	13
2022	24	23
2023	62	59
2024	144	138
2025	176	170

Notes: Column 2 presents the cumulative total FDSWs across the state in a given academic year. The third column presents to total number of FDSWs in our sample after creating our matched control group (see Section 4.1 for details). Academic year is denoted by the spring term of that year; for example, 2019 is the 2018-19 academic year. We drop one district from our analysis that adopted a FDSW in 2020 but closed before the end of our sample period.

Table 2: Traditional vs. Four Day School Week Districts, AY2019

	All districts			Sample districts		
	Traditional	FDSW	Difference	Control	FDSW	Difference
<i>ACS Characteristics</i>						
LFP Rate	81.041	80.524	-0.517	81.014	80.524	-0.491
Female LFP Rate	70.932	70.225	-0.707	70.975	70.225	-0.750
Unemployment Rate	3.539	3.472	-0.067	3.833	3.472	-0.361
Median Earnings	41,760.964	37,566.677	-4,194.287***	38,512.225	37,566.677	-945.548
Working in Sales and Office	19.454	18.825	-0.629	19.632	18.825	-0.807
Working in Service	15.977	17.327	1.350	16.001	17.327	1.326
Working in Ag	6.762	8.322	1.560*	8.752	8.322	-0.430
Working in Construction	9.865	9.813	-0.052	9.774	9.813	0.039
Working in Ed, Health	25.140	25.136	-0.004	24.656	25.136	0.481
<i>District Characteristics</i>						
Rural	0.590	0.790	0.200***	0.765	0.788	0.023
N Schools	8.414	2.540	-5.875***	2.883	2.588	-0.294*
N Students	5876	947	-4929***	1096	973	-123
N Teachers	484	94	-390***	108	97	-11
Days Taught	165.018	162.840	-2.178***	163.076	163.275	0.199
<i>Teacher Characteristics</i>						
White	0.731	0.817	0.085***	0.818	0.814	-0.004
Hispanic	0.205	0.123	-0.082***	0.134	0.126	-0.008
Black	0.046	0.046	0.000	0.034	0.046	0.012*
Female	0.790	0.790	0.000	0.791	0.791	0.000
Master's Degree	0.166	0.146	-0.020***	0.144	0.143	-0.000
Standard Cert	0.895	0.877	-0.017***	0.882	0.878	-0.004
Years Experience	7.915	7.728	-0.186**	7.855	7.722	-0.133
New Teacher	0.075	0.085	0.010***	0.083	0.085	0.002
Years Tenure	5.530	5.056	-0.474***	5.284	5.111	-0.173
New Hire	0.174	0.208	0.034***	0.194	0.204	0.010
Salary	44,395	41,900	-2495***	41,668	41,922	253
Stayer	0.823	0.798	-0.025***	0.801	0.799	-0.002
Mover	0.088	0.109	0.021***	0.104	0.108	0.004
Leaver	0.089	0.094	0.004	0.096	0.093	-0.002
<i>Student Characteristics</i>						
White	0.475	0.567	0.092***	0.565	0.562	-0.003
Hispanic	0.420	0.330	-0.090***	0.345	0.335	-0.010
Black	0.064	0.067	0.003	0.054	0.067	0.013
Female	0.486	0.485	-0.002	0.487	0.485	-0.002
Acad At Risk	0.433	0.417	-0.016	0.428	0.414	-0.014
Free and Reduced Price Lunch	0.529	0.581	0.052***	0.573	0.579	0.006
Limited English Prof	0.144	0.097	-0.047***	0.100	0.100	0.000
Chronically Absent	0.098	0.099	0.002	0.104	0.098	-0.006
<i>Outcomes</i>						
Math Scores (3-8)	0.039	-0.040	-0.078***	-0.034	-0.032	0.002
Meets Math	0.453	0.409	-0.044***	0.415	0.411	-0.004
Reading Scores (3-8)	0.034	-0.035	-0.069***	-0.040	-0.032	0.008
Meets Reading	0.443	0.405	-0.039***	0.408	0.405	-0.002
Algebra 1 Scores	-0.015	-0.089	-0.075**	-0.106	-0.087	0.019
English 1 Scores	0.132	0.116	-0.015	0.106	0.118	0.012
Graduation Rate	0.790	0.758	-0.032***	0.773	0.765	-0.008
Enrolled in College	0.572	0.527	-0.045***	0.530	0.528	-0.002
Number of Districts	842	176		230	170	

Notes: *** p<0.01, ** p<0.05, * p<0.10

Table 3: Additional Results, Teacher Analysis

	Control Mean	Pre Average	ATT
Salary	41,668.223	-23.915 (45.320)	661.676 (465.797)
Panel A: Teacher Subgroups			
Stayer, <5 Years Experience	73.088	-0.228 (0.217)	2.239*** (0.743)
Stayer, 5-10 Years Experience	80.976	-0.396 (0.155)	2.363** (1.161)
Stayer, 10+ Years Experience	85.517	0.334 (0.325)	1.617* (0.834)
Stayer, Parent Age (25-40)	82.712	-0.373* (0.205)	2.742*** (0.911)
Stayer, Non-Parent Age	89.010	-0.386** (0.187)	1.145* (0.589)
Panel B: Recruitment Outcomes			
New Hire	19.432	0.041 (0.134)	-1.137* (0.594)
<i>Among New Hires...</i>			
New Teacher	42.350	0.132 (0.442)	-6.078*** (1.332)
Years Experience	3.975	0.004 (0.025)	0.428*** (0.115)
Master's Degree	12.913	0.289 (0.260)	2.833*** (0.834)
Pre-FDSW Value-Added, Math	-0.320	-0.035 (0.026)	0.378** (0.164)
Pre-FDSW Value-Added, Reading	-0.349	0.004 (0.032)	-0.200 (0.165)

Notes: Column presents the AY2019 control mean for each outcome. Column 2 presents the average of pre-treatment ($t < 0$) estimates. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Additional Results, Student Analysis

	Control Mean	Pre Average	ATT
Panel A: Test Scores by Level			
Math Scores, Elementary	-0.173	0.001 (0.004)	-0.017 (0.018)
Meets Math, Elementary	42.154	-0.040 (0.157)	-0.508 (0.777)
Reading Scores, Elementary	-0.125	-0.003 (0.003)	0.012 (0.017)
Meets Reading, Elementary	40.628	-0.120 (0.117)	0.403 (0.721)
Math Scores, Middle	0.104	0.001 (0.006)	0.002 (0.018)
Meets Math, Middle	41.041	-0.010 (0.220)	0.415 (0.854)
Reading Scores, Middle	0.044	-0.004 (0.003)	-0.022 (0.016)
Meets Reading, Middle	40.981	-0.127 (0.120)	-1.113 (0.817)
Algebra 1 Scores	-0.106	-0.001 (0.007)	-0.020 (0.029)
English 1 Scores	0.106	-0.004 (0.005)	-0.009 (0.019)
Panel B: Additional Outcomes			
Enrolled in Same District	86.799	0.052 (0.060)	-0.258 (0.286)
Student-Teacher Ratio	9.207	0.001 (0.016)	0.004 (0.078)
Chronically Absent	10.383	-0.080 (0.085)	-0.054 (0.606)
Has Discip Action	10.798	-0.009 (0.072)	-0.489 (0.318)
Number of Discip Actions	2.068	-0.021* (0.011)	-0.053 (0.041)

Notes: Column presents the AY2019 control mean for each outcome. Column 2 presents the average of pre-treatment ($t < 0$) estimates. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Low Minutes vs. Other Districts

	Other	Low Minutes	Difference
<i>District Characteristics</i>			
Rural	0.794	0.694	-0.100
N Schools	2.773	2.755	-0.018
N Students	1087	1057	-30
N Teachers	107	105	-2
Average Days Taught	163.497	163.517	0.020
<i>Teacher Characteristics</i>			
White	0.819	0.783	-0.037
Hispanic	0.126	0.138	0.011
Black	0.040	0.065	0.025*
Female	0.788	0.790	0.002
Master's Degree	0.145	0.140	-0.005
Standard Cert	0.879	0.866	-0.014
Years Experience	7.761	7.678	-0.084
Share New Teacher	0.082	0.086	0.004
Years Tenure	5.140	5.046	-0.095
Share New Hire	0.199	0.211	0.012
Salary	42,037	41,847	-191
Share Stayer	0.800	0.790	-0.010
Share Mover	0.112	0.110	-0.001
Share Leaver	0.088	0.100	0.011
<i>Student Characteristics</i>			
White	0.566	0.508	-0.059
Hispanic	0.334	0.367	0.032
Black	0.062	0.093	0.031
Female	0.482	0.486	0.004
Free and Reduced Price Lunch	0.577	0.576	0.000
Limited English Prof	0.105	0.105	0.001
Chronically Absent	0.097	0.110	0.013
<i>Student Outcomes</i>			
Overall Math (3-8)	-0.043	-0.034	0.008
Overall Reading (3-8)	-0.046	-0.050	-0.004
Meets Math	0.402	0.413	0.010
Meets Reading	0.395	0.402	0.007
Algebra 1	-0.089	-0.087	0.002
English 1	0.111	0.100	-0.011
HS Grad Rate	0.758	0.780	0.021
First Yr College Enr	0.531	0.521	-0.010
Number of Districts	97	49	

Notes: *** p<0.01, ** p<0.05, * p<0.10

Table 6: Additional Results, Student Outcomes

	Control Mean	Low Minutes		Other	
		Pre Average	ATT	Pre Average	ATT
Days Taught	163.076	-0.316 (0.212)	-11.503*** (1.278)	-0.092 (0.104)	-14.473*** (0.984)
Monthly Minutes	8,173.270	-79.954 (61.386)	-1624.947*** (239.266)	42.108 (27.000)	414.043 (260.438)
Avg Math Scores, Elem	-0.173	0.000 (0.009)	-0.004 (0.029)	0.002 (0.004)	-0.016 (0.023)
% Meets Math, Elem	42.154	-0.219 (0.296)	-0.737 (1.054)	0.046 (0.164)	-0.193 (1.060)
Avg Reading Scores, Elem	-0.125	-0.004 (0.005)	-0.011 (0.014)	-0.002 (0.003)	0.033 (0.024)
% Meets Reading, Elem	40.628	-0.183 (0.189)	-1.577** (0.727)	-0.108 (0.129)	1.847** (0.915)
Avg Math Scores, Mid	0.104	-0.001 (0.009)	-0.069*** (0.020)	0.000 (0.006)	0.038 (0.024)
% Meets Math, Mid	41.041	-0.102 (0.365)	-3.011*** (0.877)	0.018 (0.230)	2.099* (1.195)
Avg Reading Scores, Mid	0.044	-0.001 (0.004)	-0.078*** (0.027)	-0.005 (0.003)	0.006 (0.016)
% Meets Reading, Mid	40.981	0.020 (0.160)	-4.290*** (1.410)	-0.191 (0.140)	0.564 (0.783)

Notes: Column 1 presents the AY2019 control mean for each outcome. Standard errors in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Struggling vs. Other Districts

	Other	Struggling	Difference
<i>District Characteristics</i>			
Rural	0.770	0.857	0.087
N Schools	2.681	2.229	-0.453
N Students	1031	752	-278
N Teachers	103	73	-30*
Days Taught	163.540	162.253	-1.287
<i>Teacher Characteristics</i>			
White	0.836	0.728	-0.108***
Hispanic	0.117	0.160	0.043
Black	0.033	0.096	0.063***
Female	0.796	0.774	-0.022*
Master's Degree	0.146	0.131	-0.015
Standard Cert	0.885	0.850	-0.035***
Years Experience	7.827	7.316	-0.511**
New Teacher	0.081	0.098	0.016*
Years Tenure	5.235	4.632	-0.604***
New Hire	0.196	0.234	0.037**
Salary	41,828	42,281	453
Stayer	0.805	0.777	-0.028*
Mover	0.104	0.121	0.017
Leaver	0.091	0.102	0.011
<i>Student Characteristics</i>			
White	0.595	0.436	-0.159***
Hispanic	0.317	0.405	0.087*
Black	0.051	0.129	0.077***
Female	0.484	0.486	0.002
Free and Reduced Price Lunch	0.565	0.631	0.066**
Limited English Prof	0.094	0.124	0.031
Chronically Absent	0.091	0.127	0.036***
<i>Student Outcomes</i>			
Math Scores (3-8)	0.018	-0.227	-0.245***
Meets Math	0.437	0.312	-0.125***
Reading Scores (3-8)	0.016	-0.219	-0.235***
Meets Reading	0.425	0.328	-0.098***
Algebra 1 Scores	-0.023	-0.355	-0.332***
English 1 Scores	0.147	-0.003	-0.150***
Graduation Rate	0.765	0.763	-0.002
Enrolled in College	0.540	0.476	-0.064**
Number of Districts	135	35	

Notes: *** p<0.01, ** p<0.05, * p<0.10

Table 8: Additional Results, Student Outcomes

	Control Mean	Struggling		Other	
		Pre Average	ATT	Pre Average	ATT
Days Taught	163.076	-0.150 (0.110)	-12.368*** (1.478)	-0.150 (0.112)	-13.672*** (0.882)
Monthly Minutes	8,173.270	-85.837** (40.108)	-252.649 (832.262)	22.588 (30.635)	-312.787 (236.573)
Avg Math Scores, Elem	-0.173	0.008 (0.008)	0.012 (0.045)	-0.000 (0.004)	-0.022 (0.019)
% Meets Math, Elem	42.154	0.018 (0.236)	0.344 (1.591)	-0.054 (0.165)	-0.652 (0.838)
Avg Reading Scores, Elem	-0.125	0.000 (0.005)	-0.005 (0.021)	-0.003 (0.003)	0.015 (0.019)
% Meets Reading, Elem	40.628	-0.105 (0.168)	-2.356* (1.343)	-0.125 (0.127)	0.872 (0.744)
Avg Math Scores, Mid	0.104	0.003 (0.006)	-0.005 (0.033)	0.000 (0.006)	0.003 (0.019)
% Meets Math, Mid	41.041	0.004 (0.230)	-0.227 (1.814)	-0.015 (0.241)	0.533 (0.924)
Avg Reading Scores, Mid	0.044	-0.010* (0.006)	-0.038 (0.029)	-0.002 (0.003)	-0.019 (0.018)
% Meets Reading, Mid	40.981	-0.374 (0.248)	-3.150** (1.393)	-0.078 (0.125)	-0.764 (0.895)

Notes: Column 1 presents the AY2019 control mean for each outcome. Standard errors in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figures

Figure 1
Texas School Districts with Four-Day School Weeks
By year of adoption

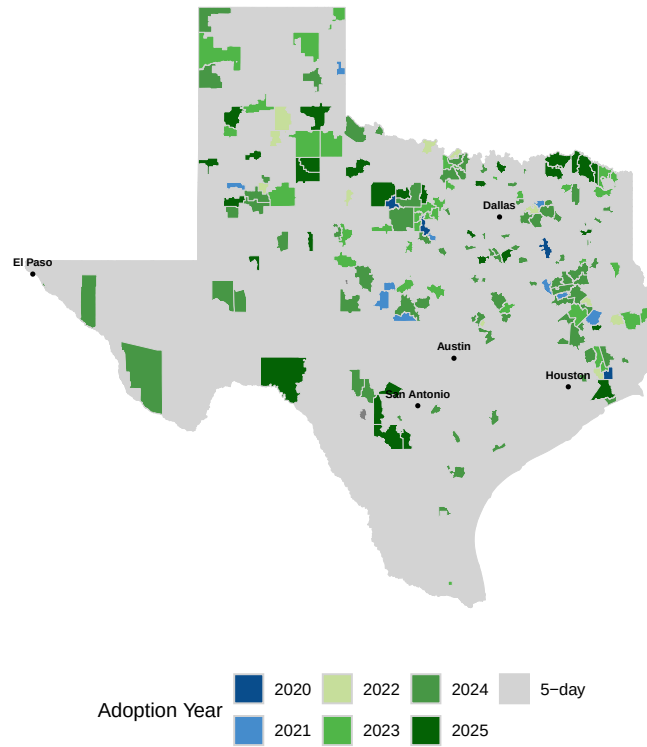


Figure 2
Four-Day School Week Districts and Matched Controls

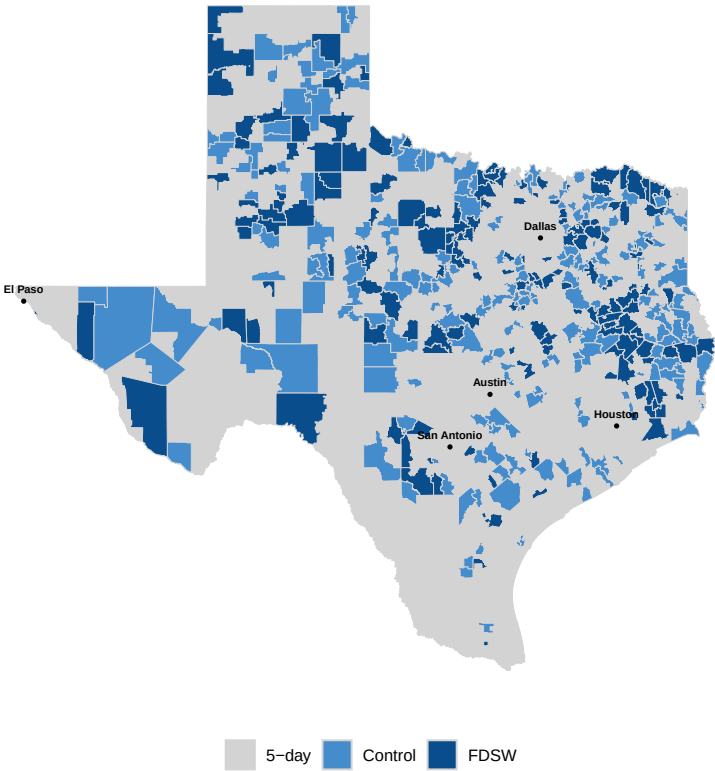


Figure 3: Pooled: Average days taught

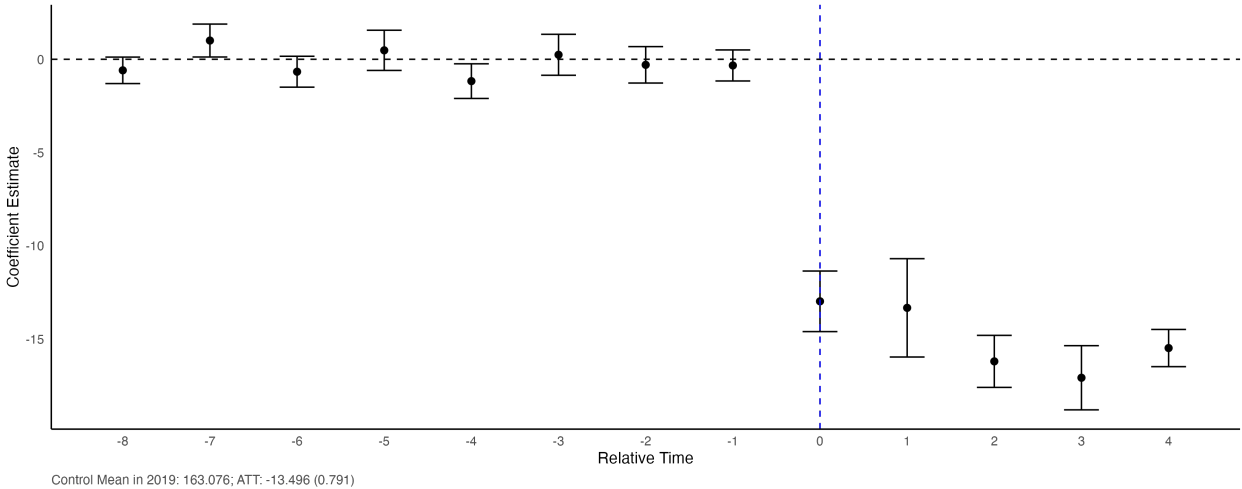


Figure 4: Pooled: Share of teachers who stayed in district

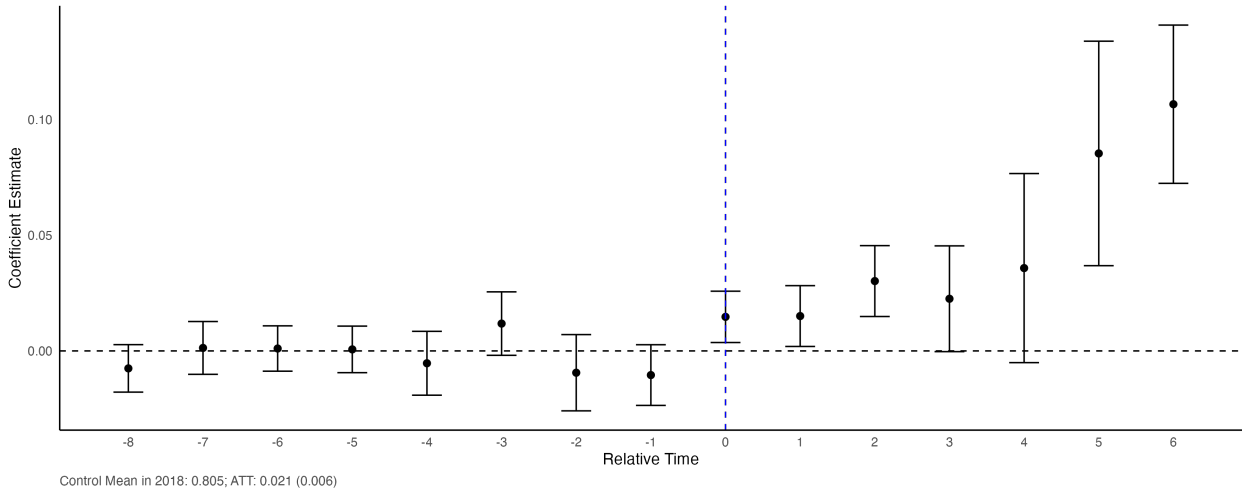


Figure 5: Pooled: Share of teachers who moved districts

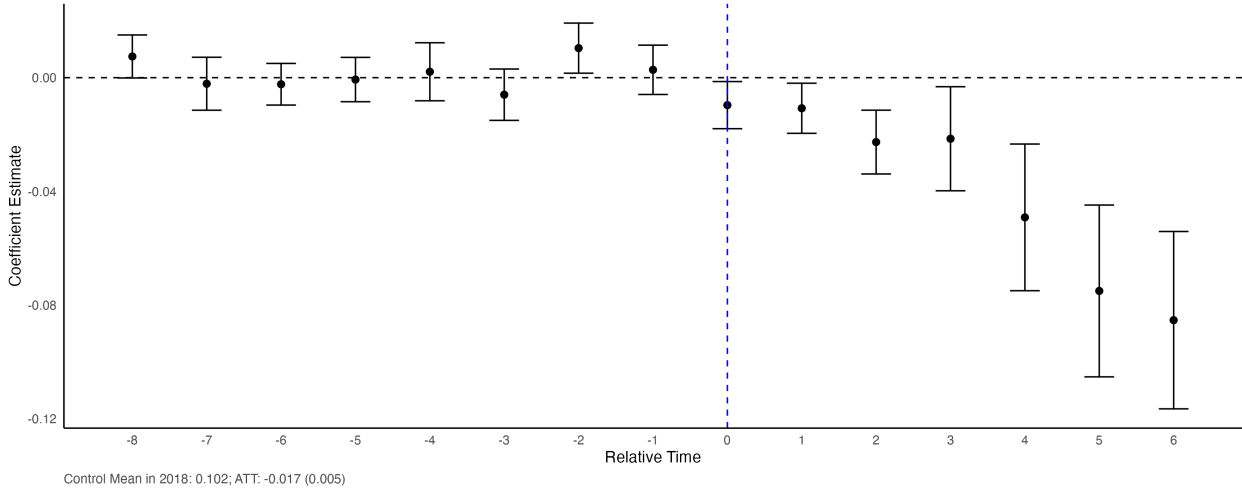


Figure 6: Pooled: Share of teachers who left teaching

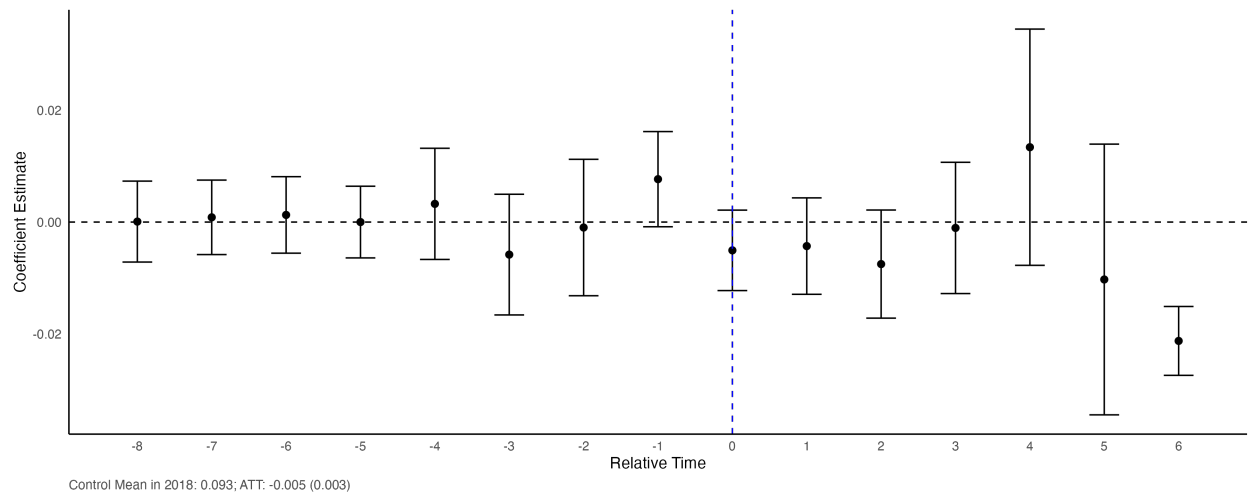


Figure 7: Pooled: Overall math scores (3rd-8th grade)

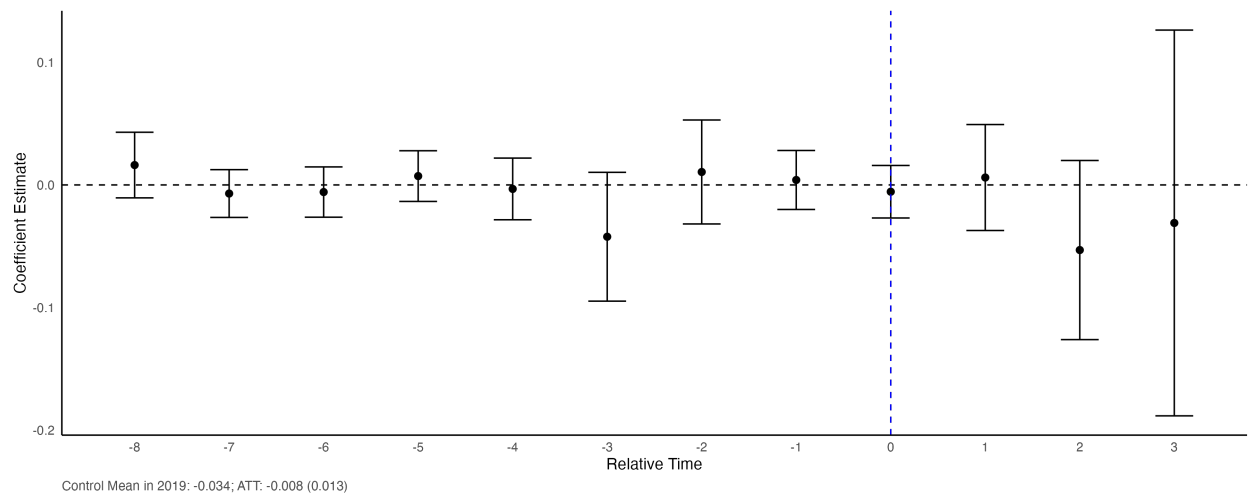


Figure 8: Pooled: Share of students who meet math standard

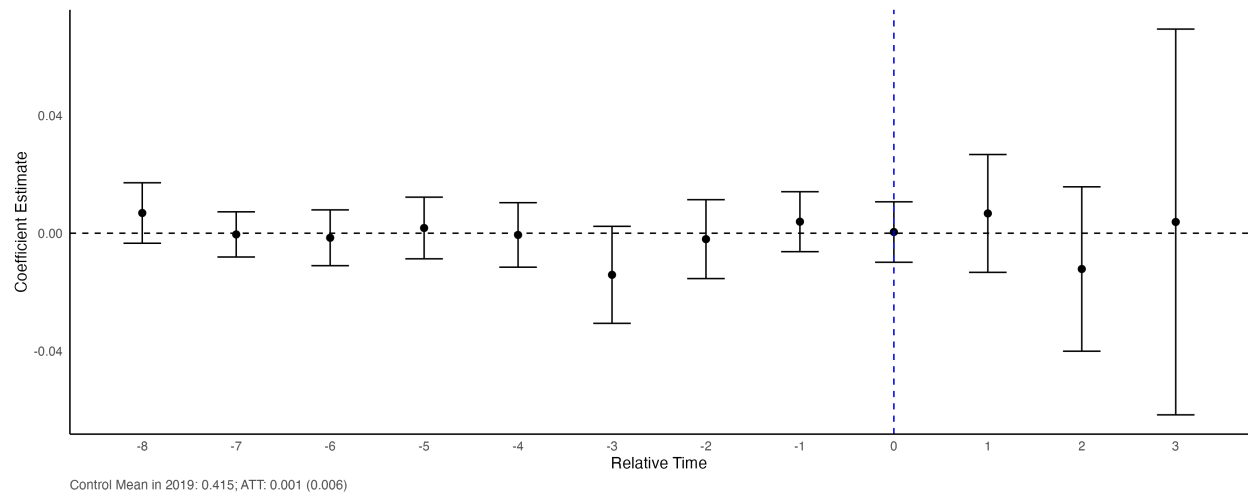


Figure 9: Pooled: Overall reading scores (3rd-8th grade)

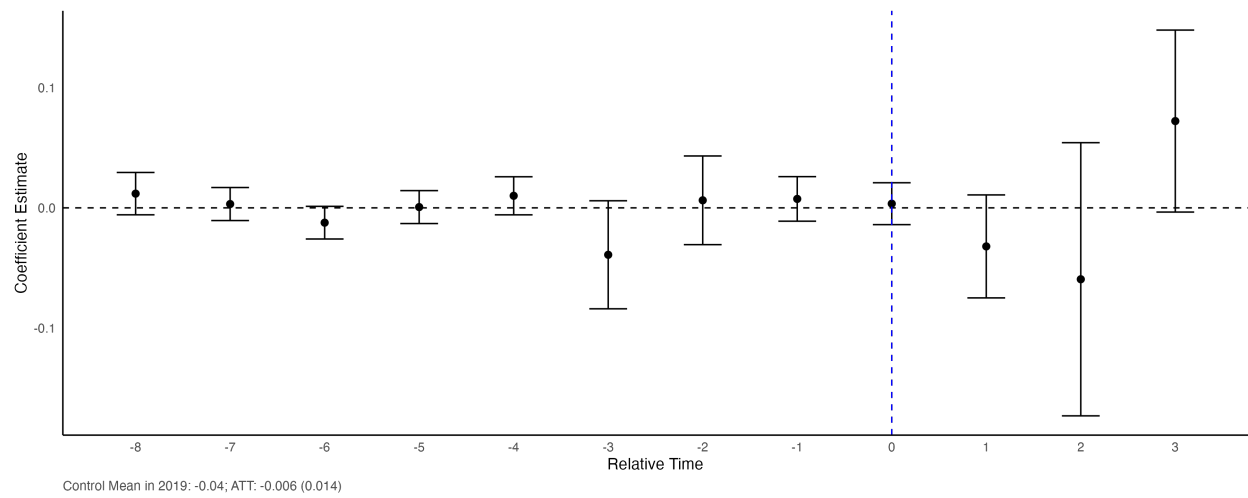


Figure 10: Pooled: Share of students who meet reading standard

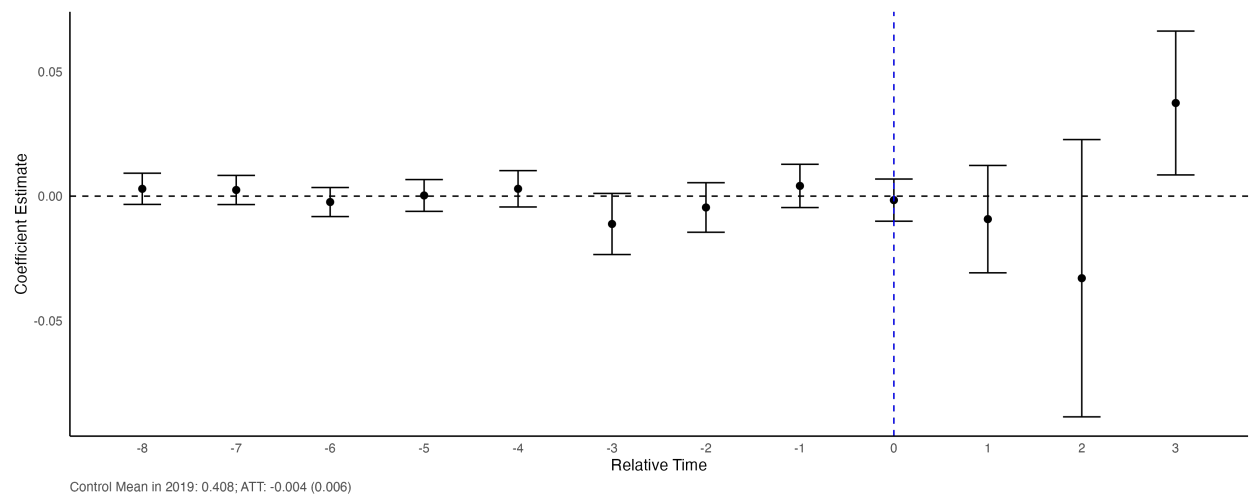


Figure 11: FDSW Effects on Math Scores

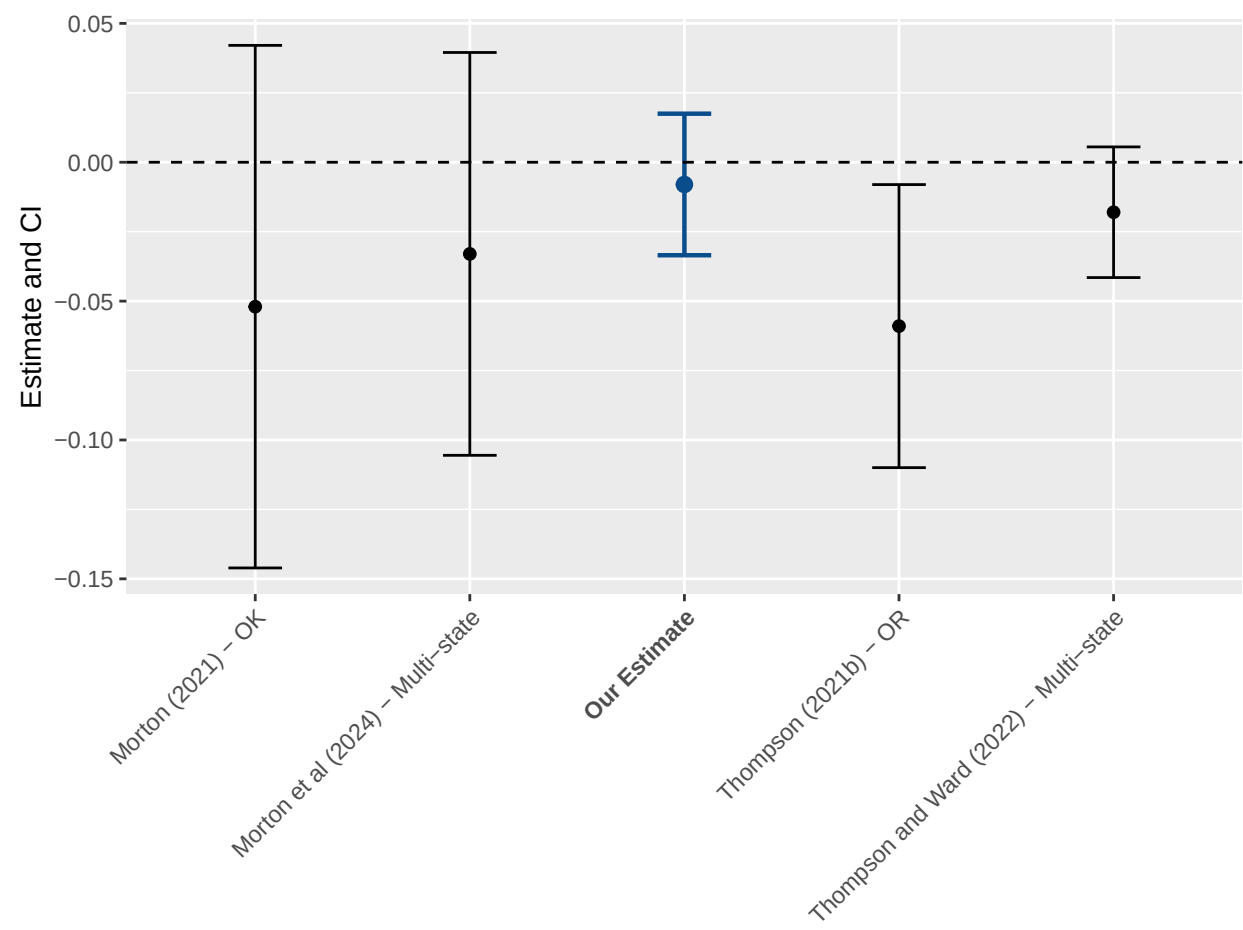


Figure 12: FDSW Effects on Reading Scores

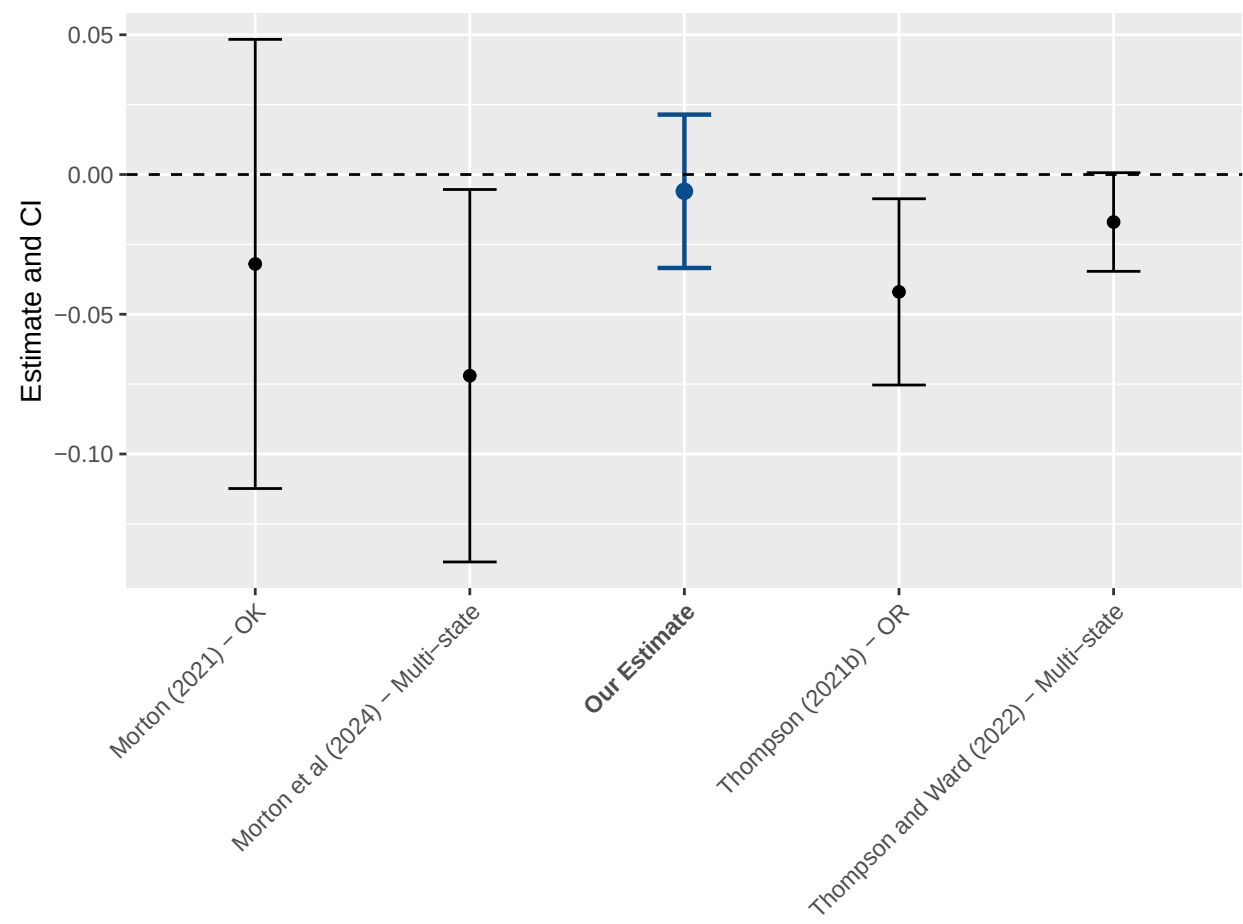


Figure 13: Pooled: Share of high school students working

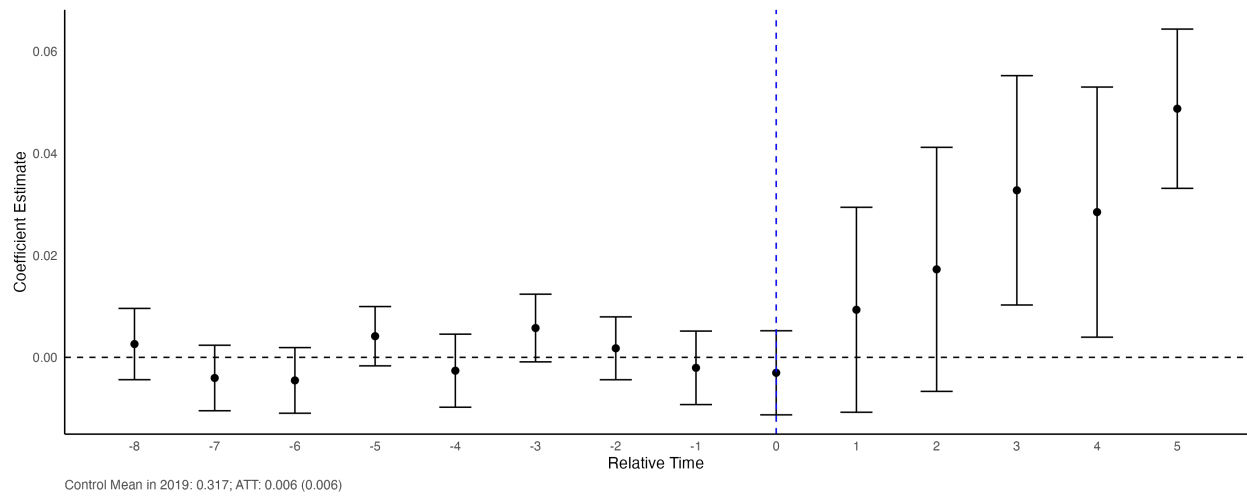


Figure 14: Pooled: Average annual earnings of high school students

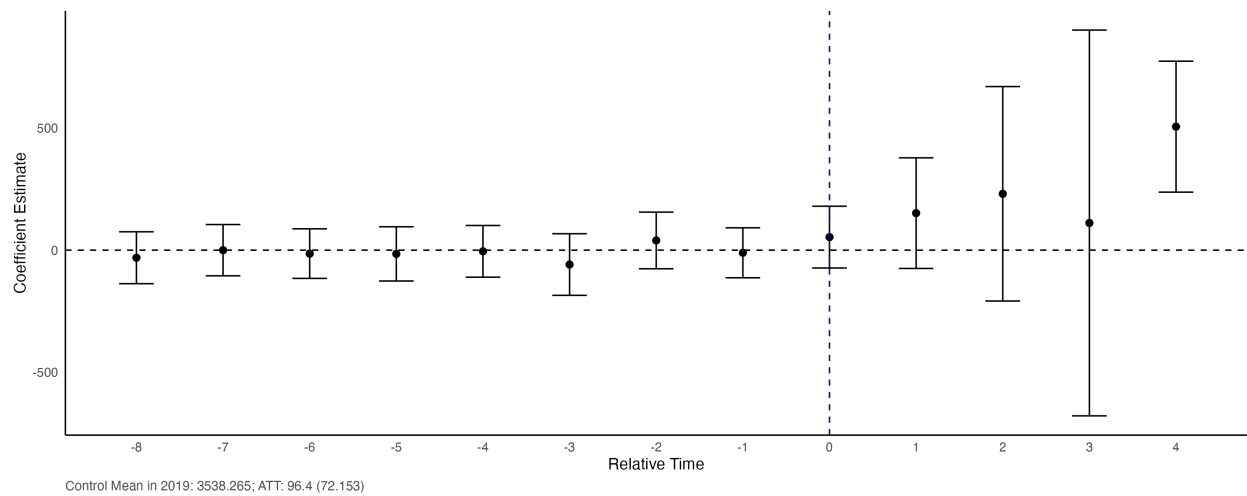


Figure 15: Pooled: Graduation rate

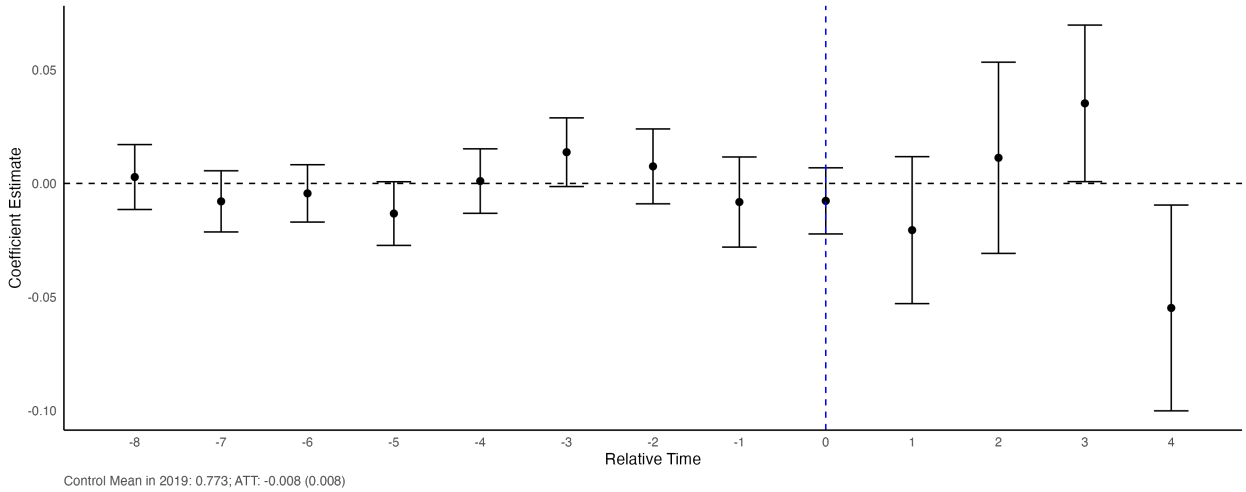


Figure 16: Pooled: Share of graduates enrolled in college

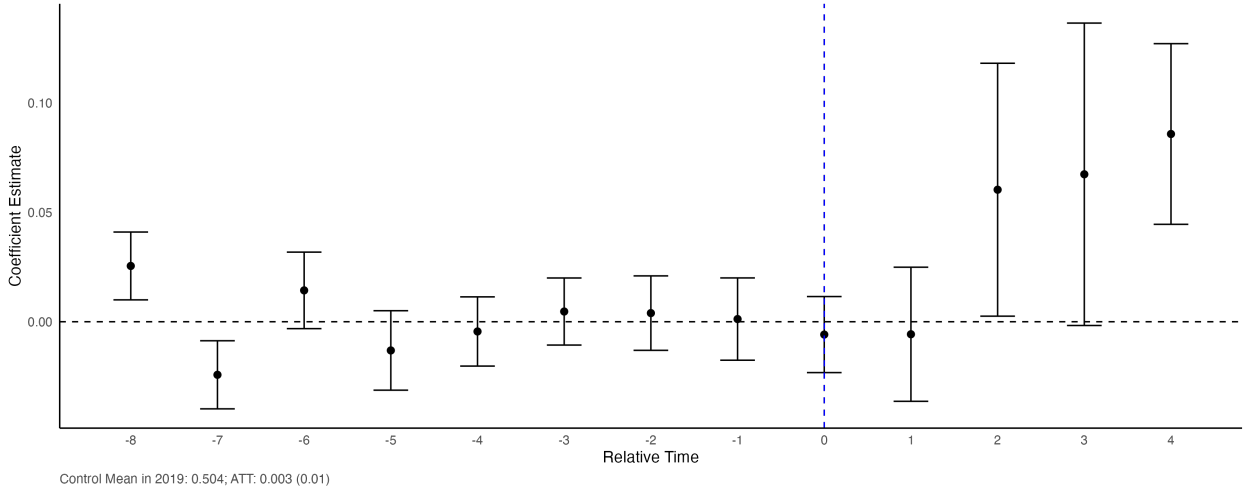


Figure 17: By Minutes: Overall math scores (3rd-8th grade)

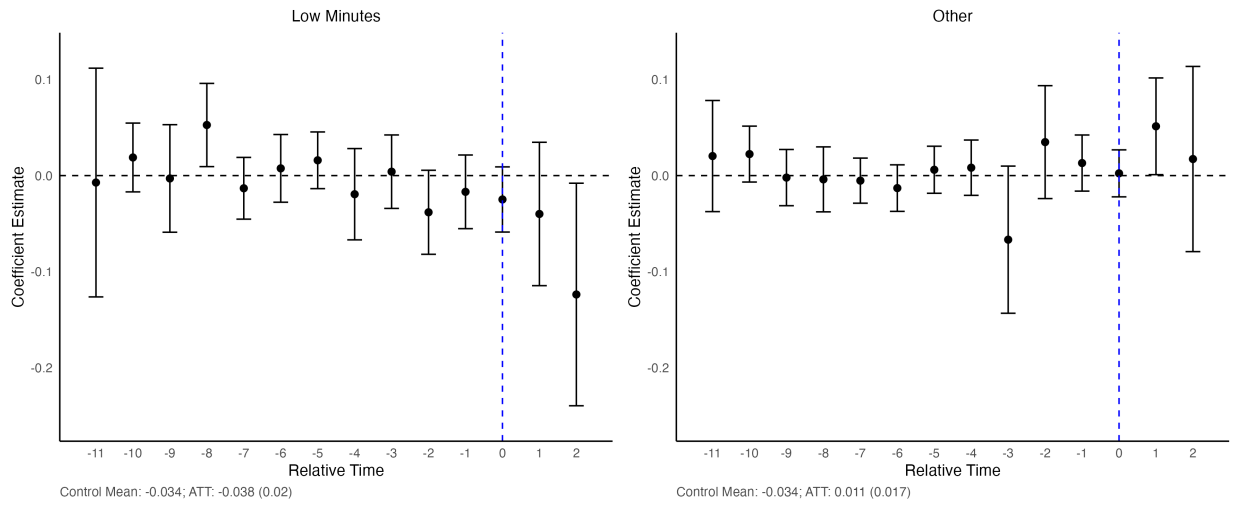


Figure 18: By Minutes: Share of students who meet math standard

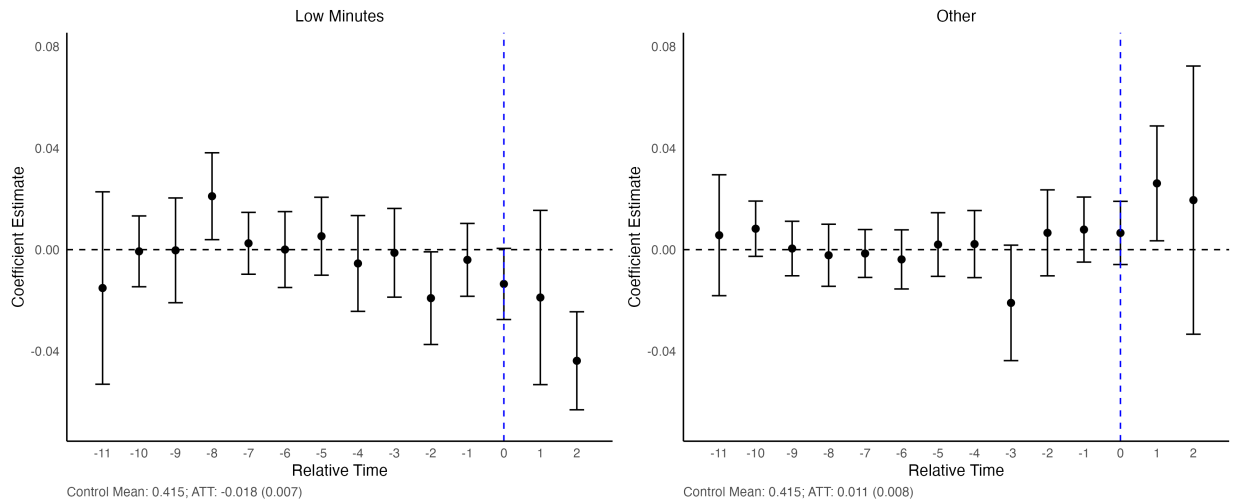


Figure 19: By Minutes: Overall reading scores (3rd-8th grade)

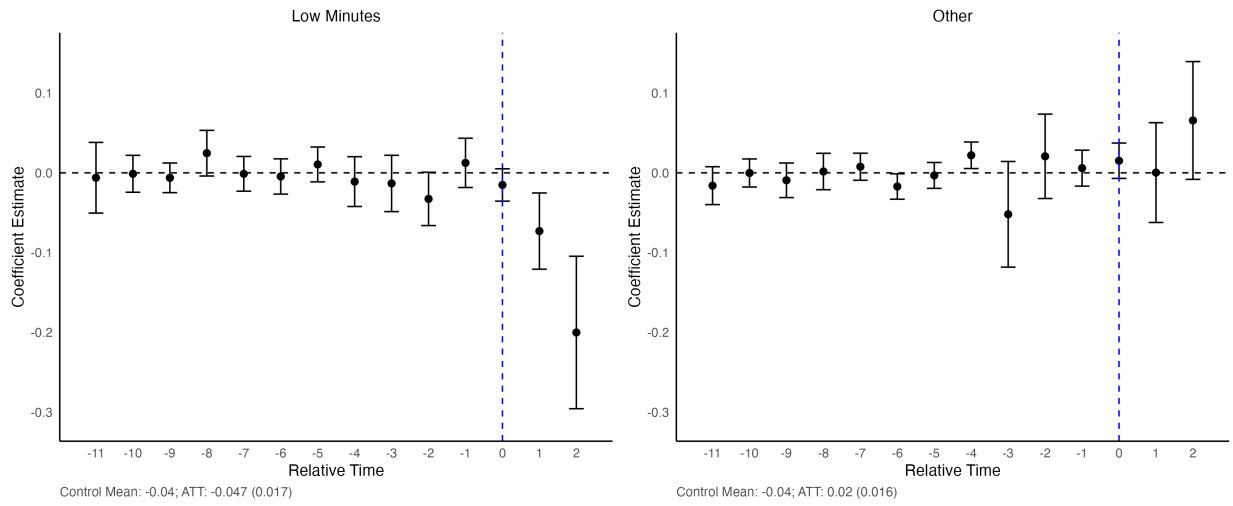


Figure 20: By Minutes: Share of students who meet reading standard

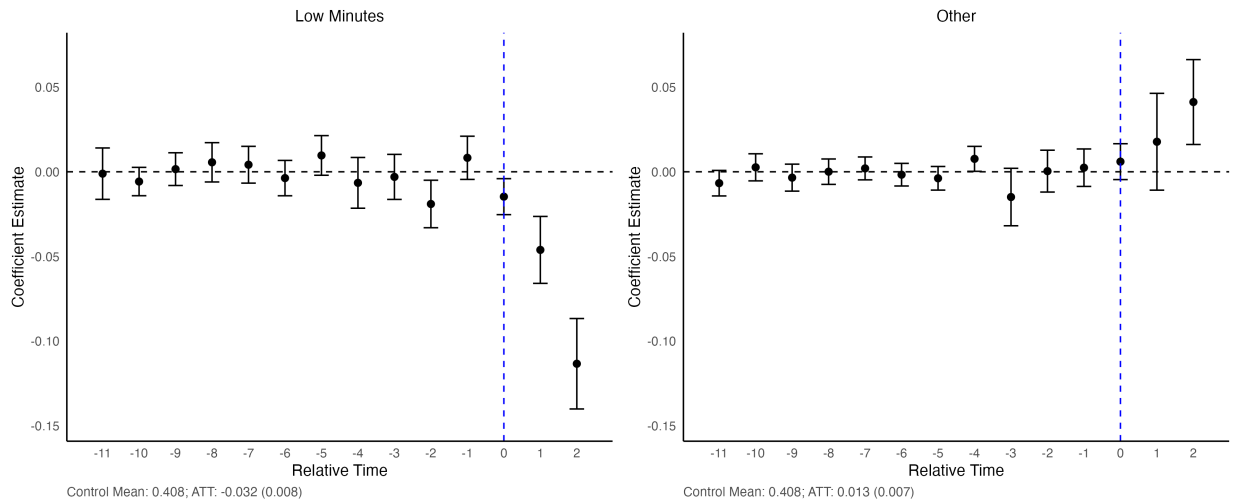


Figure 21: By Accountability Score: Overall math scores (3rd-8th grade)

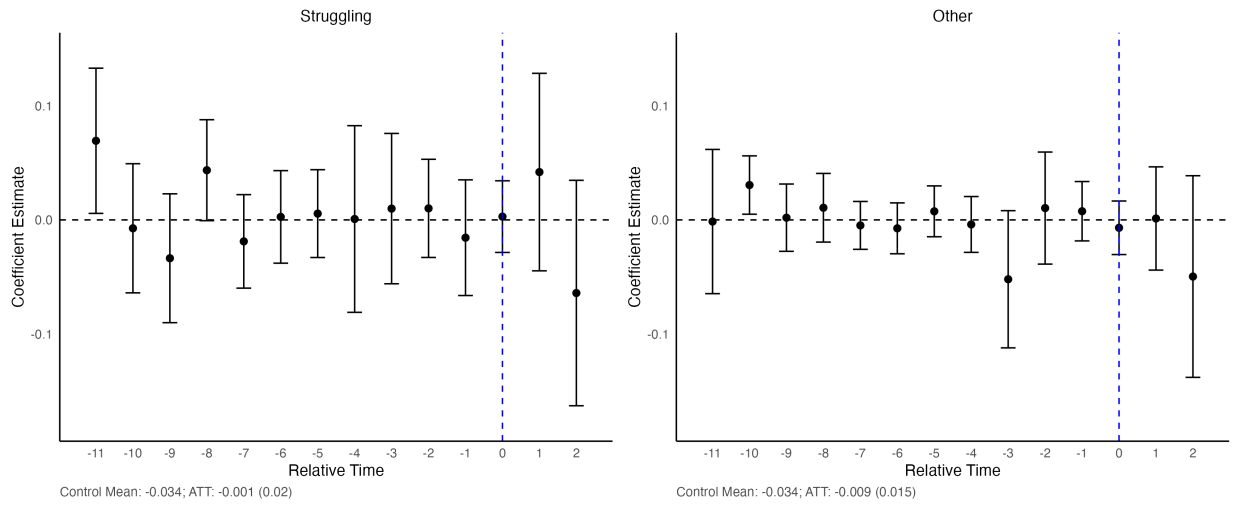


Figure 22: By Accountability Score: Share of students who meet math standard

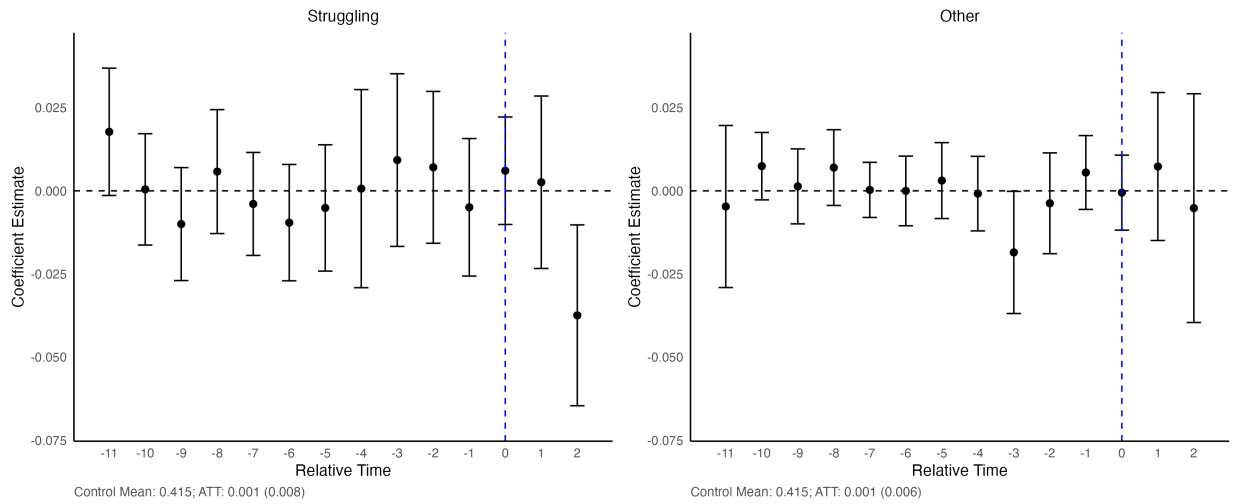


Figure 23: By Accountability Score: Overall reading scores (3rd-8th grade)

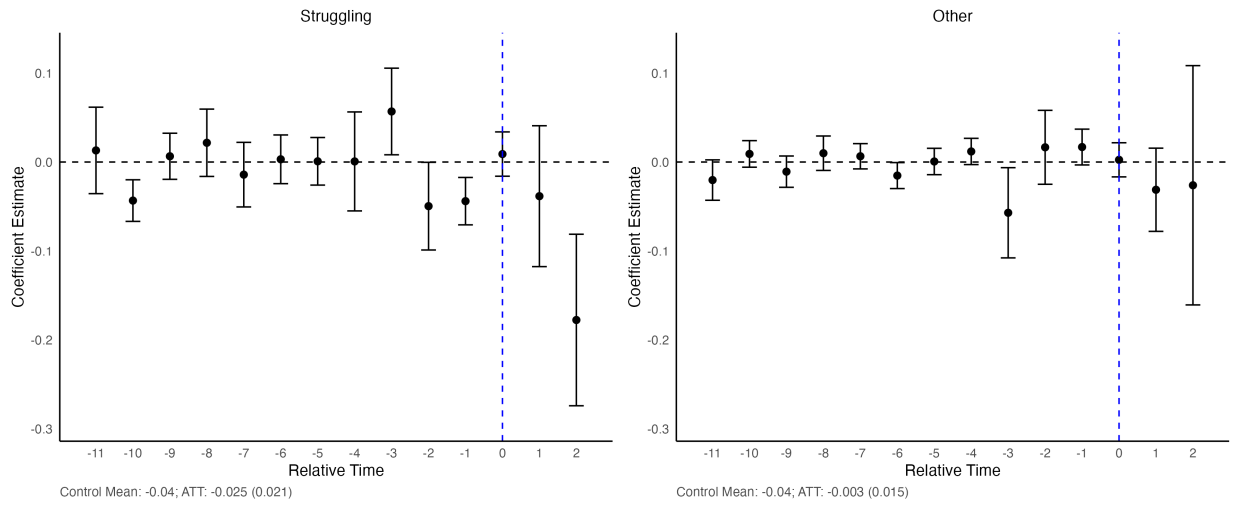
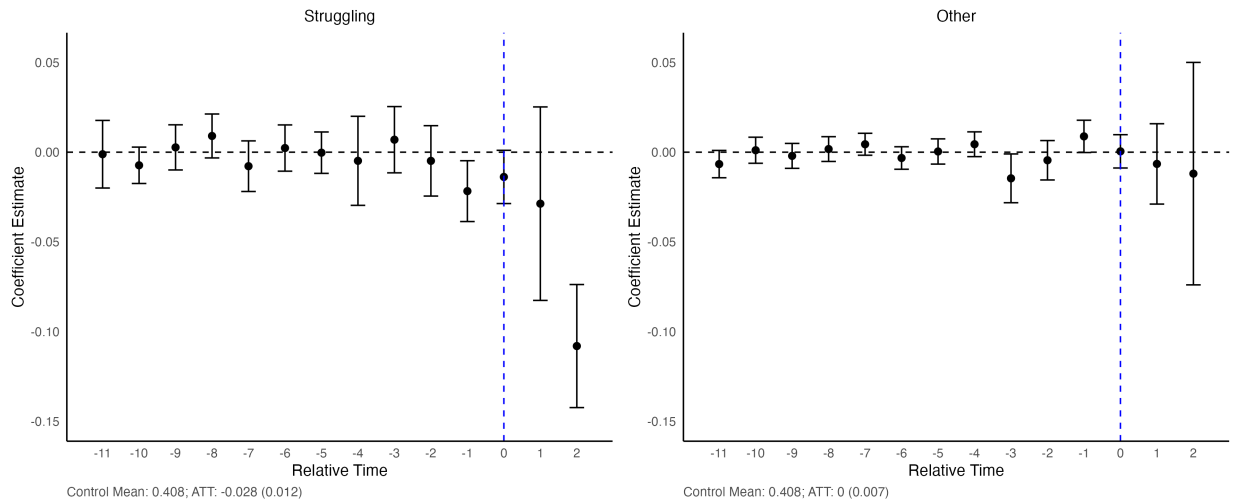


Figure 24: By Accountability Score: Share of students who meet reading standard



Appendix

A Heterogeneity: By Adoption

In this appendix, we examine whether effects differ by the timing of FDSW adoption. This analysis is motivated by patterns in our pooled event-study plots that appear related to the nature of using an unbalanced panel. Coefficients at later relative-time horizons are estimated using only early-adopting cohorts, and we observed that pooled event-study estimates shifted noticeably once later relative-time periods were driven primarily by districts that adopted before AY2022. We therefore split the sample into early adopters (districts that adopted a FDSW before 2022) and later adopters (districts that adopted in 2022 or later) as a robustness check on our pooled results.

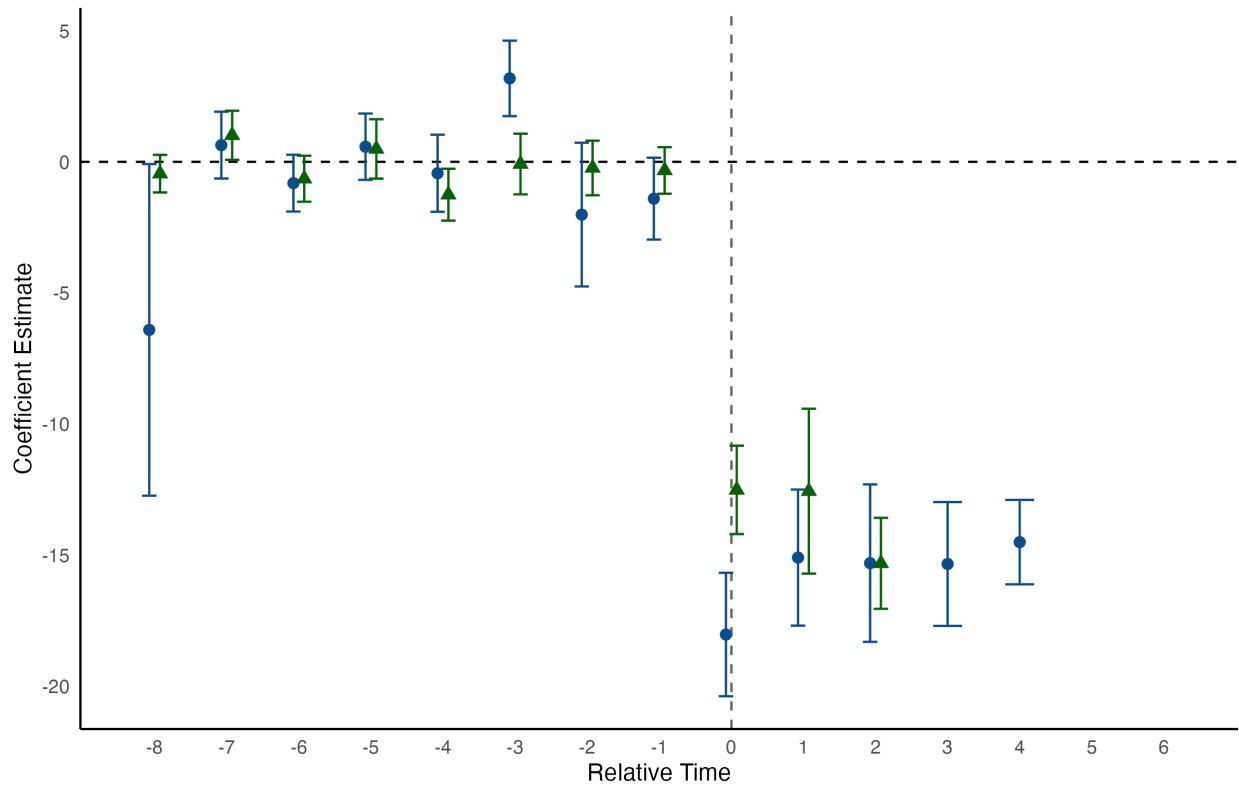
For this analysis, we match early and later adopters separately to control districts using the same process described in Section 4.1. Though 15 districts adopted a FDSW before 2022, we drop one district that closed before the end of our sample period; of the remaining 14, 13 matched to 24 control districts. Of the remaining districts that adopted a FDSW in AY2022 or later, we were able to match 157 districts to 218 control districts.¹⁹ With only 13 early adopters, the results for this group tend to be noisier than for the later adopters; however, we find the results by early and later adopters to be distinct enough to present separately. For each group, we compare outcomes to the control districts that matched to those districts.

Among early adopters, FDSW adoption increases the share of stayers by 6.8 percentage points, or 8.5% from the control mean (Figure A.2). These results are driven by a -6.0 percentage point change in the share of teachers who move to another district (57.7% decrease from the control mean); we do not find effects on the share of teachers who leave the profession. Figures A.3 and A.4 present the event studies for movers and leavers, respectively. Among later adopters, we find a small but significant increase in the share of teachers who stay in a district by 1.6 percentage points and observe a decrease in movers of 1.1 percentage points.

As seen in Figure A.5, we find no effect on math scores or proficiency for students in early-adopting districts (Figure A.6). Among later adopters, we find a significant negative effect two years post adoption ($r = 2$) on both math scores and proficiency. Additional years of data are needed to determine whether this result is attributable to the cohort of districts adopting in 2022 or if this effect exists across cohorts. Figure A.7 presents results for reading scores by early and later adopters. FDSW adoption had no effect on reading scores in early adopters, but a positive effect of 2.4 percentage points on reading proficiency (Figure A.8). Later FDSWs experienced worsening effects on reading performance and proficiency over time, with significant negative estimates two years after adoption ($r = 2$).

¹⁹Note that a control district can match to both an early and later adopter and thus be included in both analyses.

Figure A.1: By Adoption: Average days taught



• Early FDSW Control Mean in 2019: 162.235; ATT: -15.771 (1.136)
 ▲ Later FDSW Control Mean in 2021: 158.588; ATT: -12.643 (0.849)

Figure A.2: By Adoption: Share of teachers who stayed in district

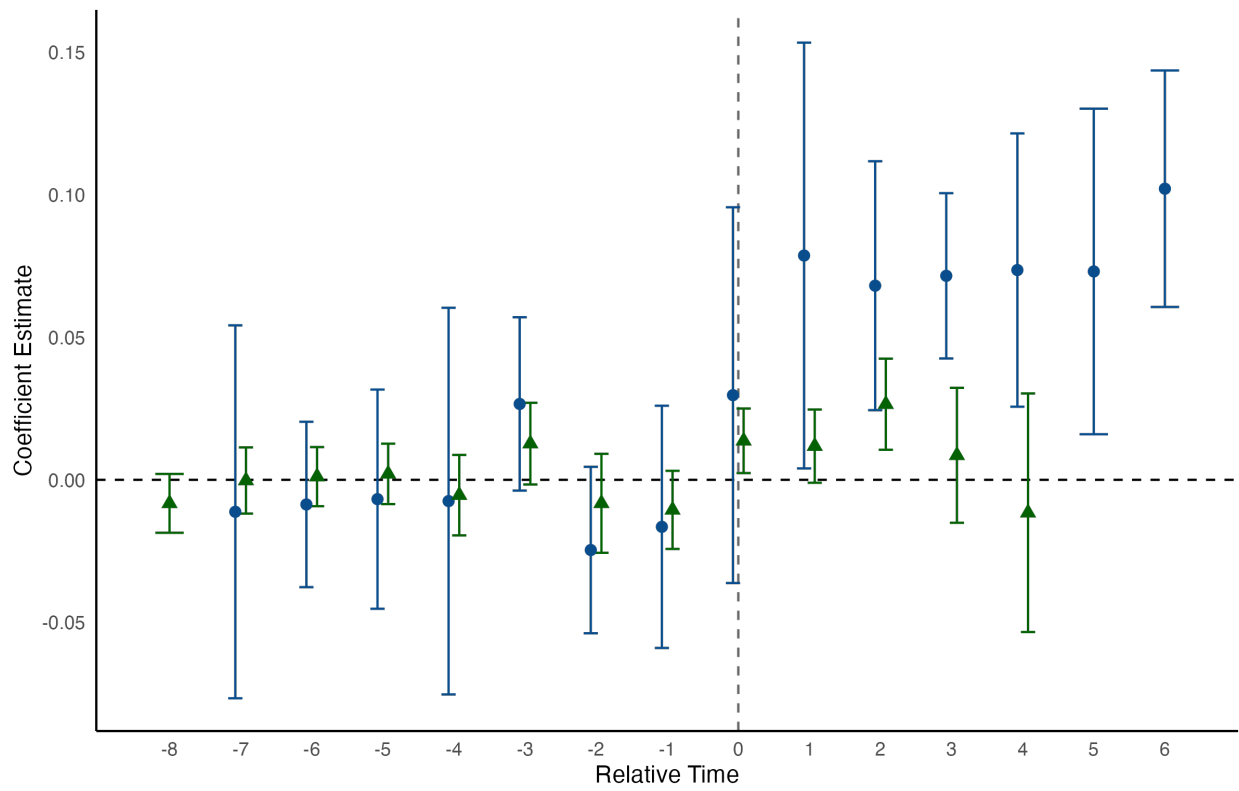
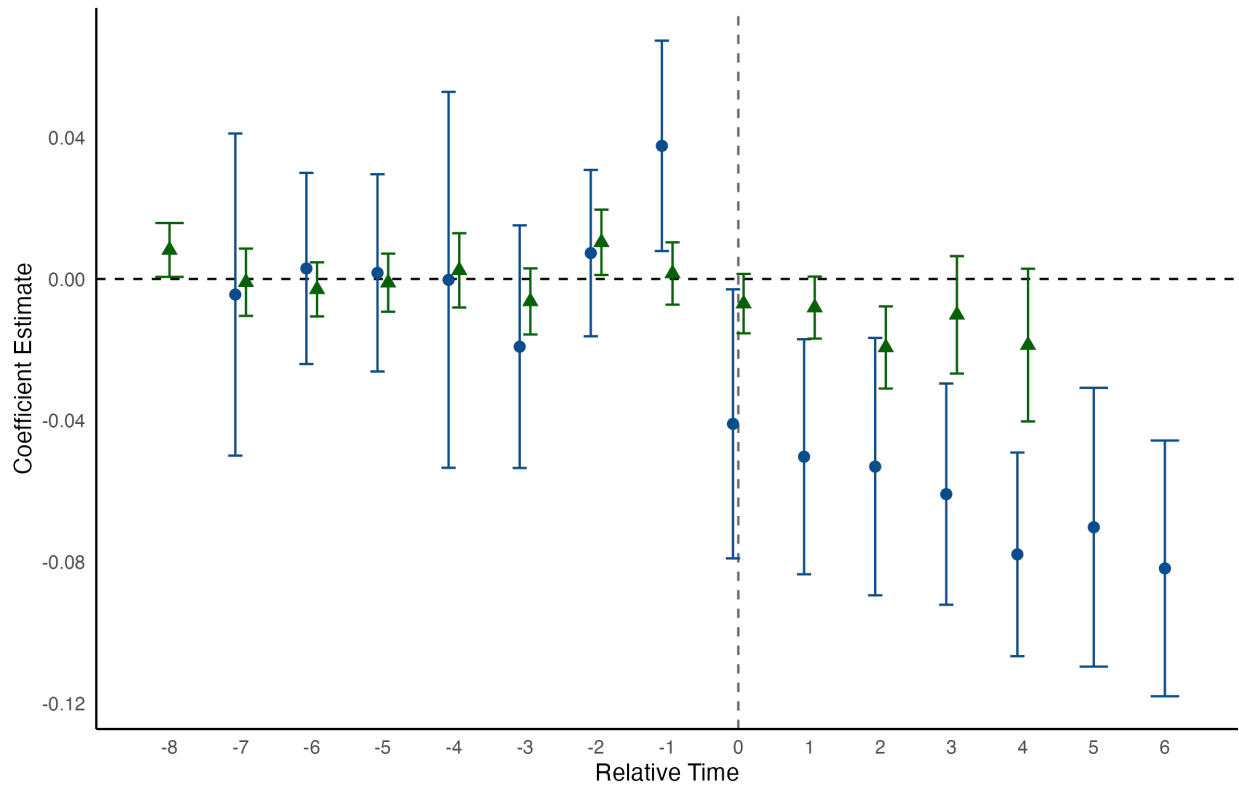


Figure A.3: By Adoption: Share of teachers who moved districts



● Early FDSW Control Mean in 2018: 0.104; ATT: -0.06 (0.015)
▲ Later FDSW Control Mean in 2020: 0.056; ATT: -0.011 (0.004)

Figure A.4: By Adoption: Share of teachers who left teaching

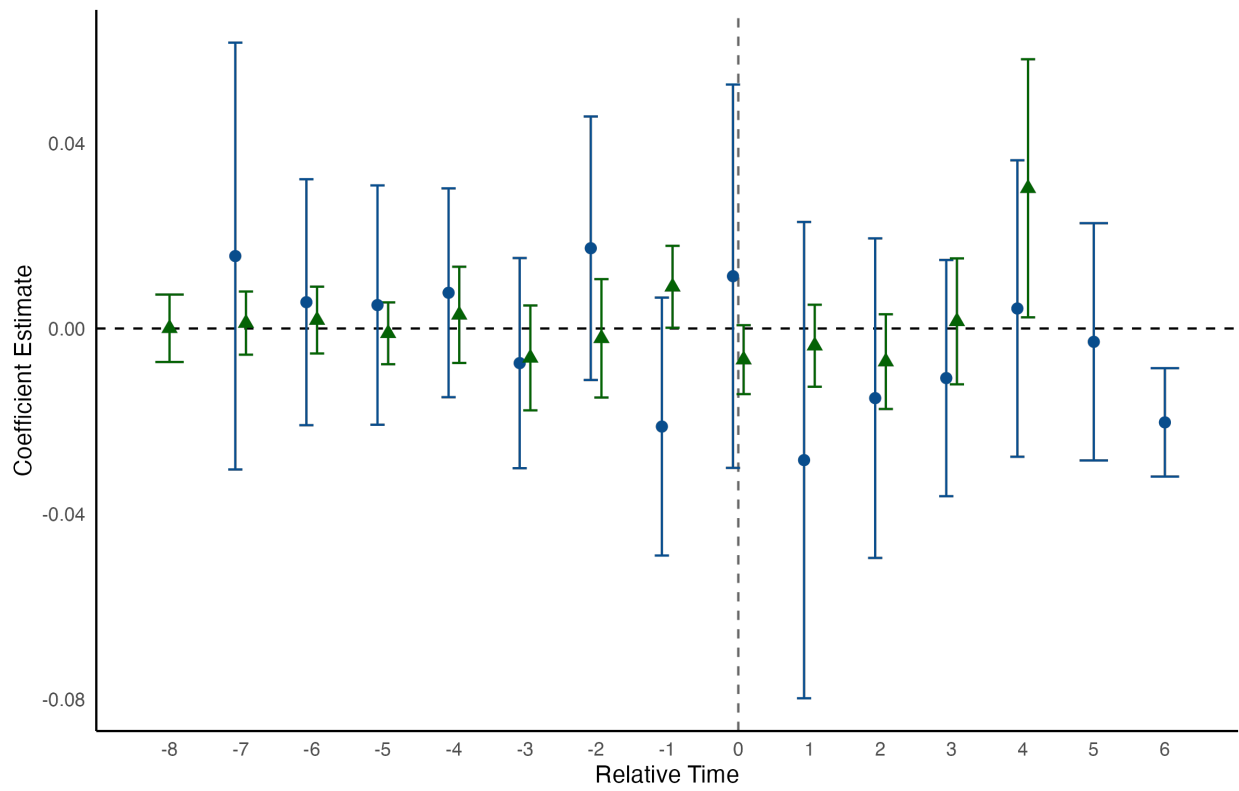
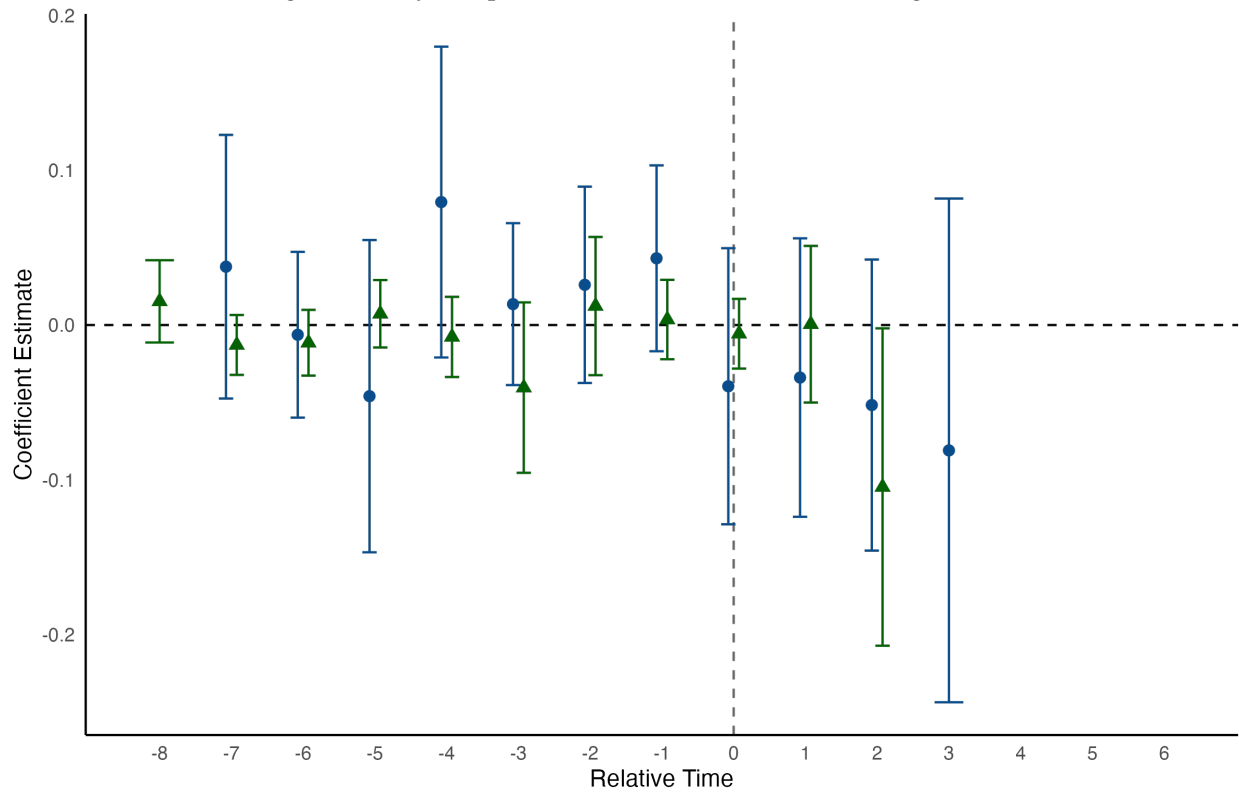
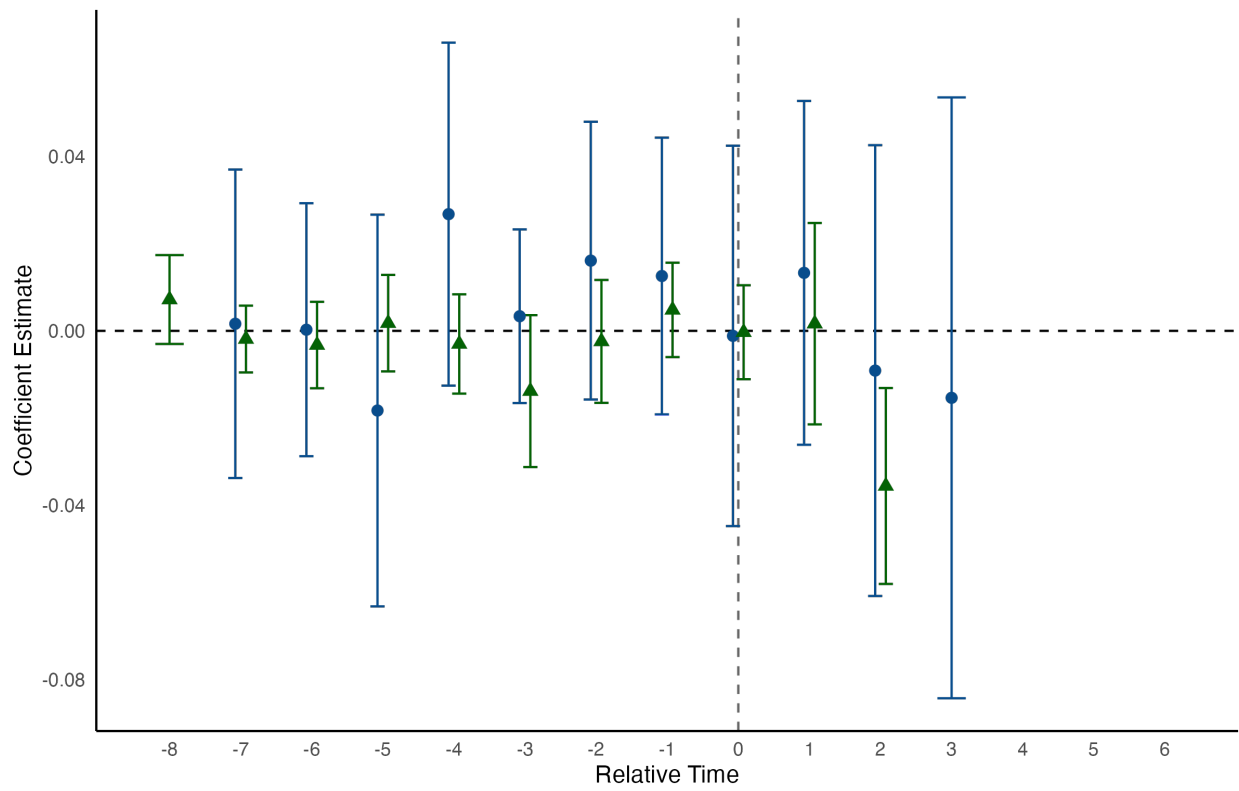


Figure A.5: By Adoption: Overall math scores (3rd-8th grade)



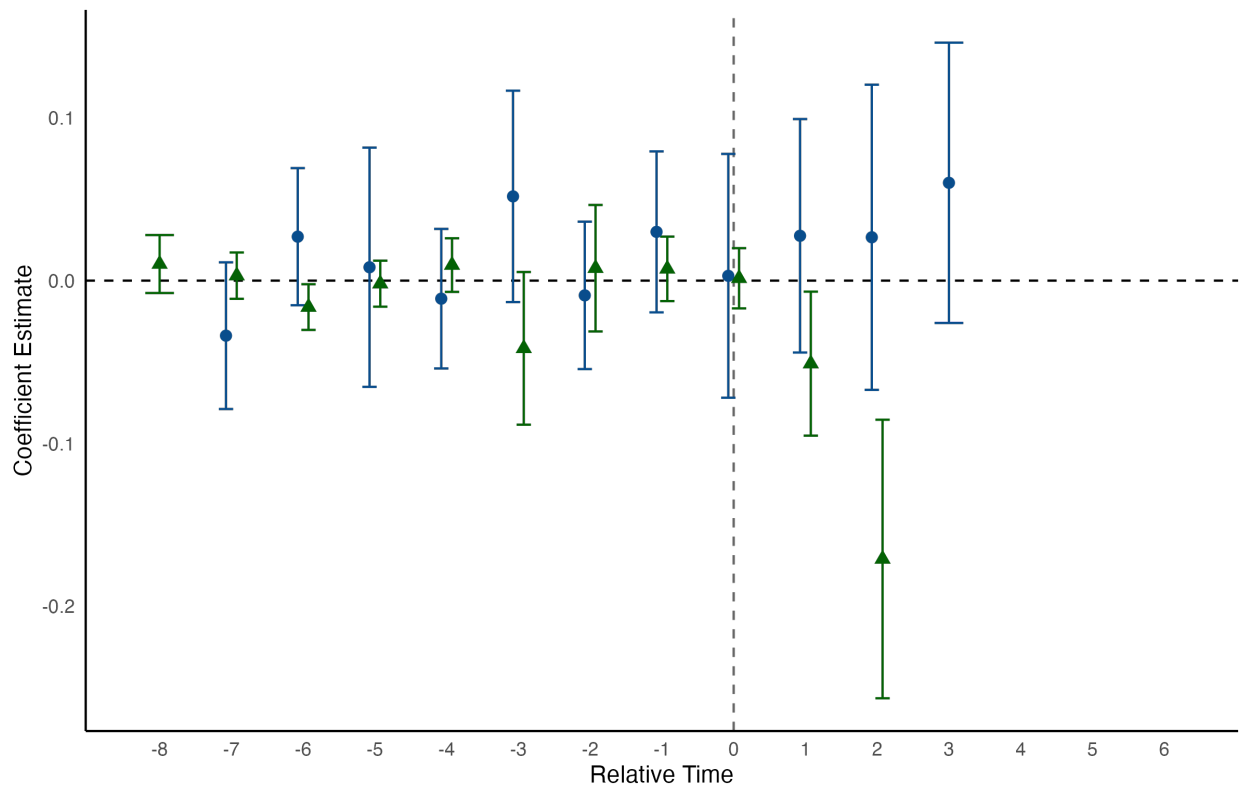
● Early FDSW Control Mean in 2019: -0.097; ATT: -0.052 (0.051)
 ▲ Later FDSW Control Mean in 2021: 0.291; ATT: -0.009 (0.013)

Figure A.6: By Adoption: Share of students who meet math standard



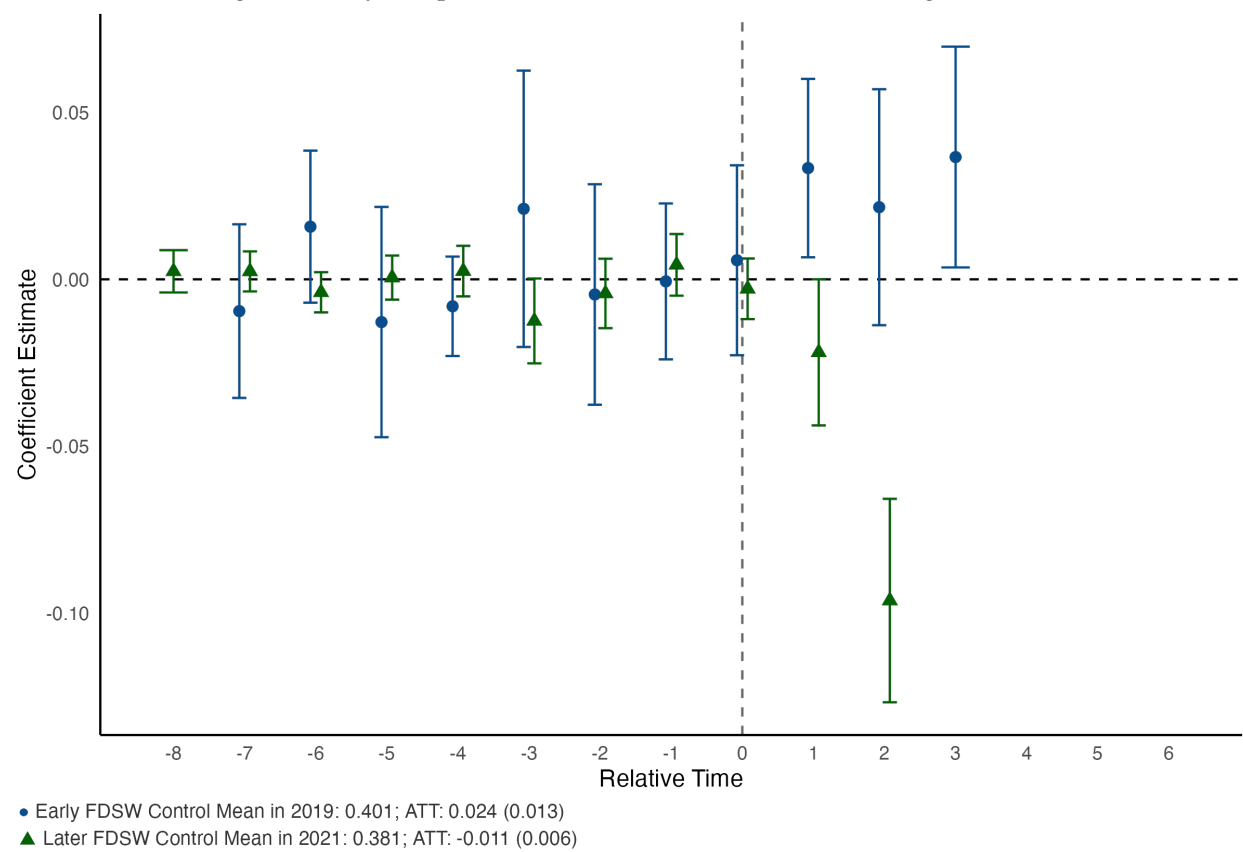
• Early FDSW Control Mean in 2019: 0.392; ATT: -0.003 (0.024)
 ▲ Later FDSW Control Mean in 2021: 0.341; ATT: -0.002 (0.006)

Figure A.7: By Adoption: Overall reading scores (3rd-8th grade)



● Early FDSW Control Mean in 2019: -0.057; ATT: 0.029 (0.039)
▲ Later FDSW Control Mean in 2021: 0.265; ATT: -0.016 (0.013)

Figure A.8: By Adoption: Share of students who meet reading standard



B Robustness: No Neighbors

In this appendix, we exclude neighboring districts from the control group as a robustness check. To identify this control group, we conduct the same matching process described in Section 4.1 but restrict the set of possible control districts to those that do not have a FDSW district neighboring them. Figure B.1 shows the map of FDSW districts and non-neighboring matched control districts. The moving cost is now higher for a teacher to leave a control district for a FDSW district. The increased distance between treated and control districts reduces concerns that FDSW adoption has spillover effects on teacher turnover in control districts.

We run the same staggered DiD model as in our pooled results using this new set of non-neighboring control districts. Figure B.2 presents the event study plot for the share of teachers who stayed in the district. We find an average treatment effect of 1.8 percentage points, which is within the margin of error of our estimate using the original matched control group (2.1 p.p.). We find an ATT of -1.9 percentage points for the share of movers (compared with -1.7 p.p. in the original analysis) and 0.1 percentage points for the share of leavers (compared with 0.5 p.p. in the original analysis). The estimated effects of FDSW adoption on these teacher retention outcomes are within the margin of error of our main results when using non-neighboring control districts. In particular, the decline in the share of teachers moving to another district — the primary mechanism behind our retention finding — is nearly identical across the two specifications suggesting that improved retention following FDSW adoption is not overstated due to teachers moving from neighboring control districts into treated districts. Taken together, these findings support the credibility of our main retention results and suggest that spillovers to geographically close control districts are unlikely to be a major source of bias in our staggered DiD estimates.

Figure B.1
Four-Day School Week Districts and Non-Neighboring Controls

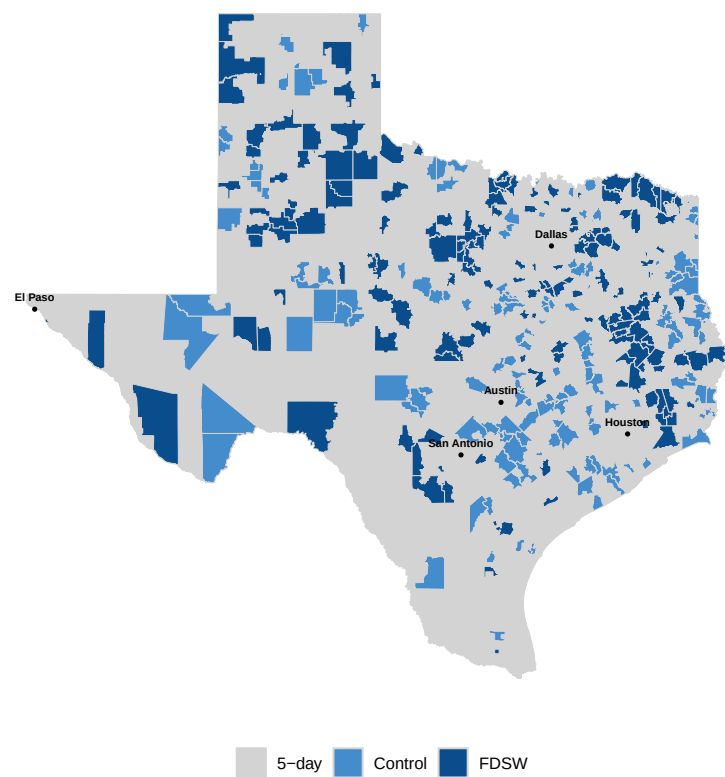


Figure B.2: Share of teachers who stayed in district

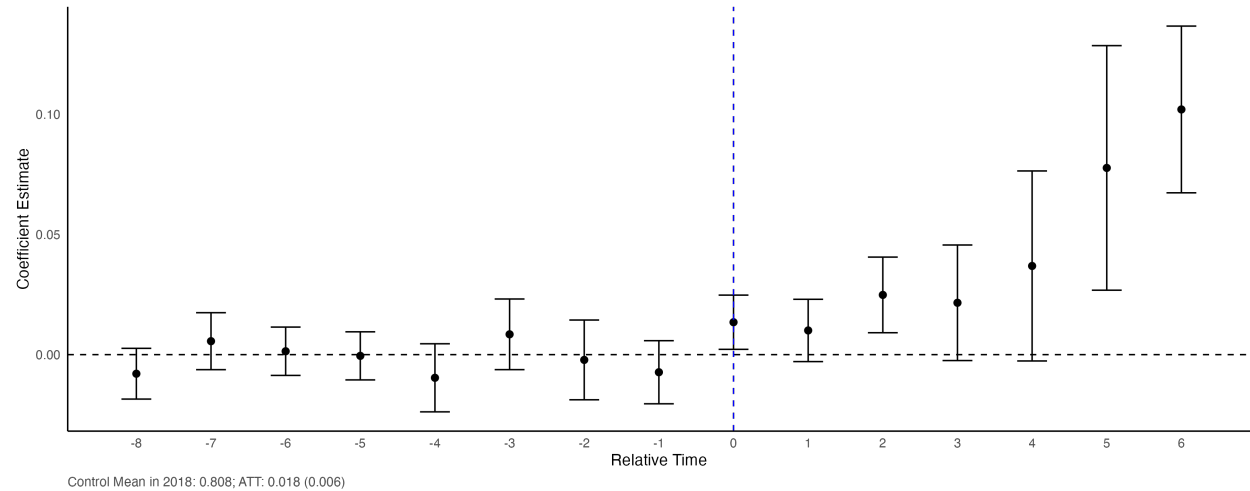


Figure B.3: Share of teachers who moved districts

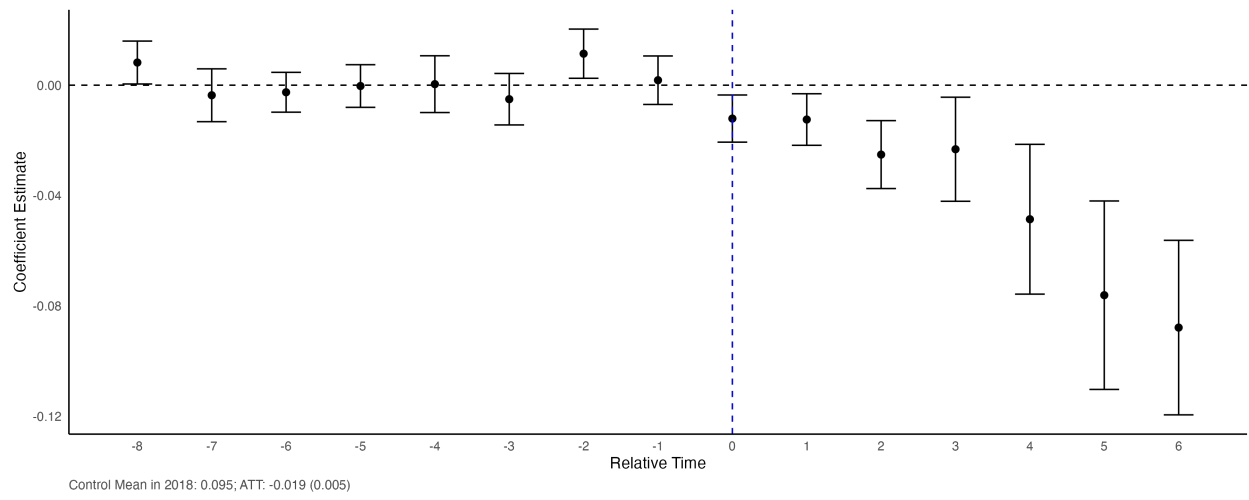


Figure B.4: Share of teachers who left teaching

